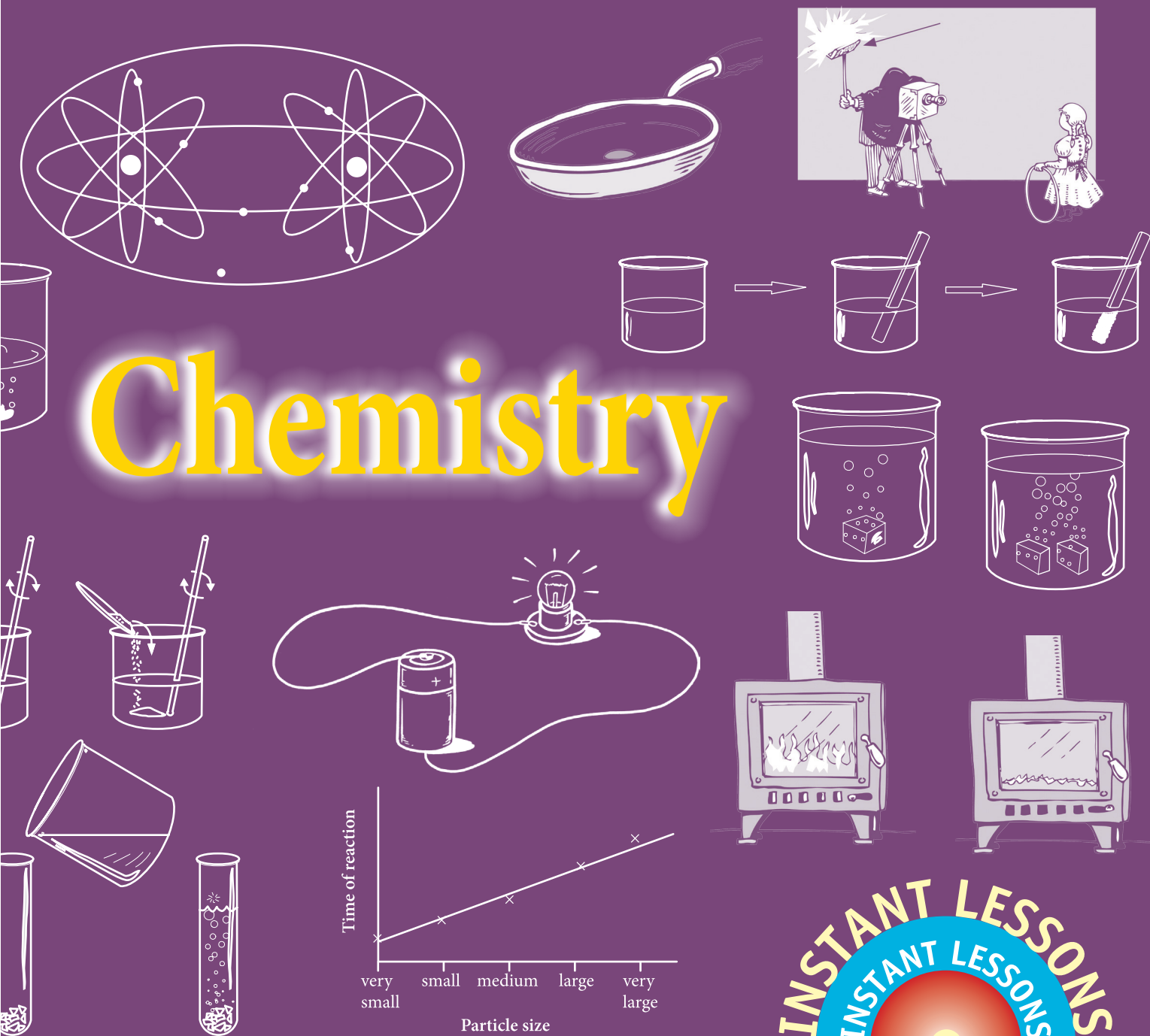


MULTIPLE CHOICE TEST TOPICS

1005 of ready-to-use topic questions!



Chemistry



Grahame Dobinson

**M U L T I P L E
C H O I C E
T E S T T O P I C S**

1005 of ready-to-use topic questions!

Chemistry

Grahame Dobinson

This book was commissioned by Jennifer Gough, production was by Piper Print Media, and David Dickson supplied the illustrations.

Grahame Dobinson is a science teacher with many years of experience in teaching all branches of science. He has a genuine love of this subject area, and is an established and successful author who has a long standing association with Blake Education.

The publisher would like to thank Peter Soucek for his assistance with the preparation of this series.

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INTRODUCTION

This series of photocopiable test topic books has been written to assist teachers to set tests and exams for students in years 9–10. The pages and questions can also be used:

- ✘ to prepare students for this specific type of assessment in more senior year levels
- ✘ as a diagnostic tool to check students' understanding of a just completed topic
- ✘ for homework, extension and revision.

The questions in this series ask students to: recall or remember a fact, make calculations, interpret data from a table, make sense of a diagram, read off values from a graph, forecast what will happen from a graph and extract information from a written passage.

Each question could be answered in the normal test manner or students could:

- ✘ be asked to justify their response
- ✘ describe why the wrong answers are incorrect
- ✘ provide examples they know that illustrate and support the correct answer.

Teachers should advise students how to record the answers – such as circling the best answer or answering on a separate answer grid. A list of helpful tips for answering multiple choice questions has been provided and was written by teachers well used to setting and marking tests and preparing their students for examinations.

An editable version of all the questions is also available on the CD-ROM that is attached to the inside back cover of this title. All material can be edited only in accordance with the user agreement.

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for answering multiple choice questions in tests and exams



Before the test or exam

- ✘ Make sure you have planned out how much time you have to answer the multiple choice section of the exam and know exactly how many marks this section is worth.
- ✘ Do as many practice multiple choice questions from past exams as you can.
- ✘ Ensure that you know the meaning of common words found in exams such as *all, many, some, sometimes, always, never, none, not, few*. Also know the definitions of key words such as *describe, analyse* and *evaluate*.

Reading test questions

- ✘ Read through the paper, and select the easiest questions to do first. Do the hardest questions last. This gives you confidence to do your best.
- ✘ Read each question and try to answer it without looking at the choices. Then read through the choices. Doing this allows you to make a clearer and more accurate choice.
- ✘ Be alert for grammatical inconsistencies between the question and the potential answers. A choice is nearly always wrong if the question and the answer don't combine to make a grammatically correct sentence.
- ✘ Don't read anything into or out of the question, or assume it has been written to trick you.
- ✘ Some questions are tricky though. For example, *Which of the following is not ...; All of the following except ...* A response may be true, but not the answer to the question set.
- ✘ The wrong answers are called 'distractors' – there's a reason for this name!

Strategies

- ✘ Eliminate answers you know aren't right and be logical in your approach.
- ✘ With the harder questions, delete the answers that are least likely to be right. Your chances of selecting a correct answer increases with every alternative you eliminate.

- ✘ If there is no penalty for guessing, and you cannot work out the right answer, make learned guesses.
- ✘ Never leave a multiple choice question unanswered.
- ✘ Don't be put off a response just because it seems too obvious or too simple.
- ✘ If two responses appear to be equally correct, eliminate the response that appears to be least related to the question being asked.
- ✘ Use a common sense approach to numerical questions. First think out whether the answer will be in 10s, 100s, 1000s or millions.
- ✘ Unless otherwise instructed, you may write your workings out or reasoning on the test paper and this will not be looked at by the marker.

At the end of the test or exam

- ✘ Leave yourself time at the end of the exam to check your answer sheet to see if you have recorded your answers correctly.
- ✘ Do not change answers unless you know that your recorded answers are incorrect.
- ✘ Do not change your answers just because there are too many or too few of a particular letter on your answer sheet.

After you have been given your marked answer sheet

- ✘ Look carefully at the questions you answered incorrectly. If you answered them wrongly because you did not understand the content, then it is 'back to the books'.
- ✘ If you made an error because your multiple choice test technique was poor, then re-read the tips for answering this kind of question and do more practice.
- ✘ If anxiety meant that you did not perform as well as you wanted, remember that sound planning and a good test technique will help you to feel more confident in any exam. Practice boosts confidence and lessens anxiety.

- 1** Modern atomic theory:
- A** is based on the idea that all substances are composed of small particles.
 - B** recognises that even atoms are composed of smaller particles.
 - C** explains why a particular amount of one chemical will react with a definite, set amount of a second chemical.
 - D** embraces all of the above ideas.
- 2** In the 1600s, before atomic theory was recognised, it was thought that:
- A** two reacting chemicals could do so regardless of the amounts of each.
 - B** the term 'atoms' only applied to some chemicals.
 - C** any elements which reacted with one another would do so in the ratio of 1:1.
 - D** only metals like silver and gold were made up of atoms.
- 3** The development of the atomic theory is usually accredited to the scientist:
- | | |
|---------------------|--------------------|
| A Lavoisier. | B Dalton. |
| C Davy. | D Einstein. |
- 4** The original proposal(s) in the early atomic theory included the idea(s) that 'atoms':
- A** were present in all substances.
 - B** could easily be divided into smaller particles.
 - C** represented the smallest 'indivisible' particles present in all matter.
 - D** had the properties of both A and C above.
- 5** One difference between early atomic theory and the more modern version is that the early version considered that atoms were:
- A** able to be 'split', whereas the modern version says that they are indivisible.
 - B** not involved in chemical reactions, while it is now known that they are.
 - C** tiny individual 'solid particles', while the modern version considers all atoms to be mainly space.
 - D** tiny individual 'solid particles', while the modern version considers atoms to be larger than originally thought.
- 6** The reason the atomic theory is called a 'theory' is that:
- A** it has more evidence to support it than a hypothesis would have.
 - B** the evidence to support it is stronger than if it were called a scientific law.
 - C** atoms have actually been seen with the aid of modern light microscopes.
 - D** there is no real evidence to support it.
- 7** Atomic theory states that:
- A** atoms of all compounds are identical.
 - B** atoms of all metallic elements are the same.
 - C** atoms of one particular element all have the same electrical charge on their nuclei.
 - D** for a particular element, the atoms of which the element is composed are all identical.

8 The atoms present in different elements are generally accepted as:

- A** being of the same size, but of different masses.
- B** having the same masses, but different sizes.
- C** having the same masses and sizes.
- D** being of different sizes and masses.

9 Modern atomic theory considers that atoms have:

- A** most of their mass towards the outside of the atoms.
- B** most of their mass concentrated in the centre of the atoms.
- C** their mass equally distributed throughout the atom.
- D** such small masses that they can be ignored in any calculations.

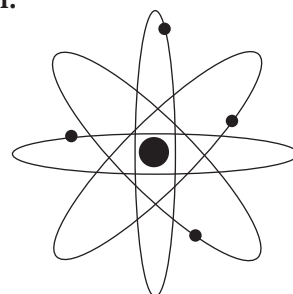
10 A current model of an atom would propose that:

- A** it is composed of particles which themselves have no electrical charge.
- B** there are only positive and negative 'sub-atomic' particles within the atom.
- C** it has a positively charged nucleus, where most of the mass of the atom lies.
- D** the majority of the mass of the atom is concentrated on the 'outer edges' of the atom.

11 Atoms of elements generally have no electrical charge. This is because:

- A** the particles of which they are composed have no charge.
- B** there are three types of charges present which balance each other out.
- C** they are nearly always in a combined state with other atoms.
- D** whatever positive charge they possess is cancelled by an equivalent amount of negative charge.

12 Atoms are often drawn as shown in the diagram.



This type of diagram shows that:

- A** sub-atomic particles appear to be orbiting a central object.
- B** there is a lot of space within the atom itself.
- C** there is a dense, central region in the atom.
- D** all of the above features are true.

13 If an atom of an element is correctly presented diagrammatically as in question 12, then it would be correct to presume that a 'particle' smaller than an atom would:

- A** occasionally be able to pass through the atom without hitting anything.
- B** always be able to pass through the atom without hitting anything.
- C** never be able to pass through the atom without hitting anything.
- D** always be absorbed by the atom and never be able to pass through.

14 The model of an atom as represented in the diagram in question 12 is referred to as:

- A** the atomic model.
- B** the nuclear model.
- C** the orbiting model.
- D** either B or C.

15 The names of the charged particles which are present in all atoms and their relative positions within the atoms are:

- A** protons in the nucleus plus neutrons in orbit around the nucleus.
- B** protons and neutrons in the nucleus plus electrons in orbit around the nucleus.
- C** protons in the nucleus and electrons in orbit around the nucleus.
- D** electrons in the nucleus and protons in orbit around the nucleus.

16 The particle called the 'neutron' is one which:

- A** adds charge to the atom, but not mass.
- B** gives mass to the atom, but not charge.
- C** explains why atoms of the one element can have different electrical charges.
- D** can have different electrical charges and is only ever present in the nucleus.

17 Most of the mass of any atom is:

- A** concentrated in the nucleus because of the electrons present.
- B** concentrated in the nucleus because the particles in it have greater mass than any particles orbiting the nucleus.
- C** equally distributed between the particles within the nucleus and the particles orbiting the nucleus.
- D** in either the nucleus or in the particles orbiting the nucleus, depending on the particular atom.

18 The particles which make up atoms are:

- A** protons which have a positive charge, electrons which also have a positive charge and neutrons which have no charge.
- B** protons which have a positive charge, electrons which have a negative charge and neutrons which also have a negative charge.
- C** protons which have no charge, electrons which have a negative charge and neutrons which have a positive charge.
- D** protons which have a positive charge, electrons which have a negative charge and neutrons which have no charge.

19 The mass of the sub-atomic particle called the electron is:

- A** so small that it would take almost two thousand of them to make up the mass of one proton.
- B** about 1/10 of that of the neutron.
- C** about equal to that of the proton and also that of the neutron.
- D** the same as that of the proton and this is why their charges counteract one another.

20 The simplest atom known is the hydrogen atom. This is due to the fact that its make-up is:

- A** nearly always a nucleus with one proton in it and one electron in orbit around the nucleus.
- B** a nucleus containing one proton and one neutron plus one electron in orbit around the nucleus.
- C** a nucleus with a single neutron in it plus one electron in orbit around the nucleus.
- D** a nucleus with a single electron in it plus one proton in orbit around the nucleus.

21 Important in the understanding of how electron(s) orbit the nucleus of an atom is the fact that:

- A** positive and negative charges attract one another.
- B** positive and negative charges repel one another.
- C** both positive and negative charges are attracted to particles with no charge.
- D** both positive and negative charges are repelled by particles with no charge.

22 In any atom of any element:

- A** the total positive charge is slightly greater than the total negative charge because of the larger size of the nucleus.
- B** the total positive charge of the nucleus is equal in size to the total negative charge of the particles orbiting the nucleus because of the effect of neutrons.
- C** the overall charge is zero.
- D** the number of neutrons is the same as the number of electrons, thus causing a zero overall charge.

23 It is possible for two atoms of any particular element to have:

- A** different masses and yet the same charge on their nuclei.
- B** different electrical charges but not different masses.
- C** the electrical charges on their nuclei different and different numbers of electrons in orbit to balance this.
- D** properties of both A and C above.

24 An atom of any element would have its chemical properties changed if:

- A** a neutron was removed from its nucleus.
- B** an electron was removed from the atom.
- C** a proton was removed from its nucleus.
- D** any of the above were to occur.

25 When two atoms of different elements become very close to one another:

- A** there is always a fairly strong bond formed.
- B** a bond may or may not be formed.
- C** a bond is formed only when the two atoms are of non-metallic elements.
- D** a bond is formed only when the two atoms are of metallic elements.

26 If two atoms of hydrogen come together and join, the resulting particle which would be formed is called a:

- A** diatom.
- B** molecule.
- C** co-atom.
- D** compound.

27 When an atom of one element combines with an atom of a different element the resulting particle:

- A** is no longer called an atom.
- B** becomes one of a compound.
- C** has statements A and B both apply to it.
- D** is still called an element.

28 When two atoms come together and 'bond' to one another, the bonding is caused by:

- A** the two nuclei attracting one another.
- B** the fact that the nucleus of each atom attracts the electrons of the other atom, since opposite charges attract.
- C** the electron 'clouds' of each atom attracting one another.
- D** some sort of interaction between the electrons of both atoms.

29 If two atoms come together and form a bond, the resulting particle would best be referred to as:

- A** a co-molecule.
- B** a diatomic compound.
- C** a diatomic molecule.
- D** a diatomic atom.

30 When two electrically neutral atoms become very close, it often occurs that one electron (or more) will basically leave one atom and start to orbit the second atom. When this occurs:

- A** the two atoms repel one another because of their different charges and so cannot form a bond.
- B** the two atoms attract one another because of their different charges and so form a bond.
- C** the two atoms become unstable and so no bond is formed.
- D** if a bond is formed, it is always a very weak one.

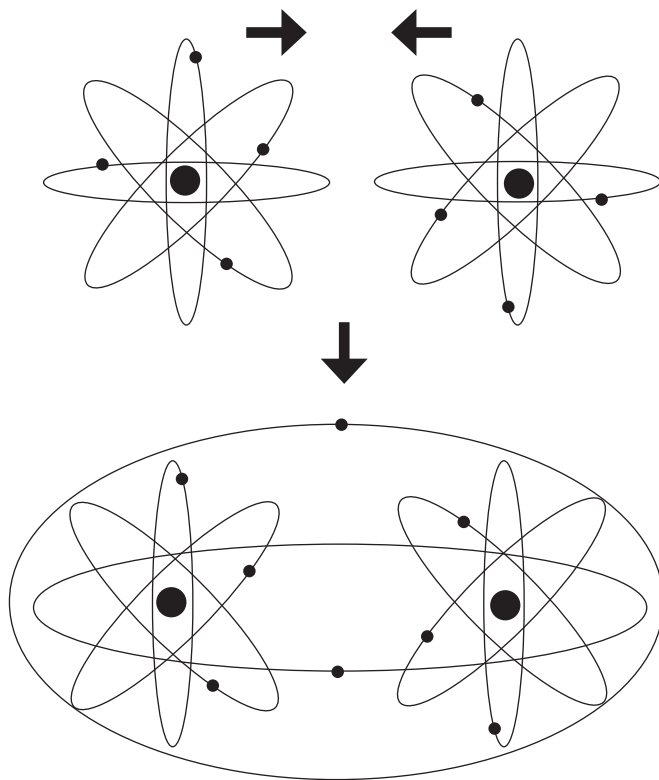
31 If a bond is formed as in question 30, then it would:

- A** not last for long as the resulting particle would be unstable.
- B** only occur when the two atoms concerned are metallic atoms.
- C** only occur when the two atoms concerned are non-metallic atoms.
- D** be called an 'ionic' bond.

32 The bond type called 'ionic':

- A** is common in compounds consisting of a metal and a non-metal.
- B** is very rare in compounds consisting of a metal and a non-metal.
- C** causes compounds which possess it to form only diatomic molecules.
- D** is a weaker type of bond and causes compounds which have it to melt easily.

33 One way by which certain atoms are able to form bonds is that, as the atoms become very close, some of the electrons start to orbit both of the atoms. This is shown below.



This type of bonding:

- A** is common with metals.
- B** is a strong one and is called a 'covalent' bond.
- C** seldom occurs between atoms of non-metals.
- D** only occurs where atoms have the same number of electrons as one another.

34 When two hydrogen atoms come together and form a molecule, the type of bond which holds them together is:

- A** always of the ionic type.
- B** sometimes ionic, but more often of the covalent type.
- C** always of the covalent type.
- D** either ionic or covalent, depending on the conditions under which they joined.

- 1** When scientists first found ways of comparing the masses of atoms, they used the mass of the hydrogen atom as a basic unit of 'one'. This was because:
 - A** using 'grams' or even 'milligrams' as units was not practical.
 - B** the hydrogen atom was the one with the smallest mass.
 - C** it meant that atoms of every other element would have masses greater than one.
 - D** all of the above statements were applicable and true.
- 2** If we say that 'a helium atom has an atomic mass of 4', then we are saying that:
 - A** each helium atom has a mass of 4.0 g.
 - B** an atom of helium has 4 times the mass of a hydrogen atom.
 - C** each atom of helium has a mass of 4.0 milligrams.
 - D** four atoms of helium have the same mass as one atom of hydrogen.
- 3** The original Periodic Table was formed as a result of scientists arranging the elements in order of increasing atomic masses and noticing that:
 - A** there was a regular (or periodic) repeat of elements which had very similar properties.
 - B** elements with names starting with the same letter had similar properties.
 - C** only metallic elements had similar properties.
 - D** only non-metallic elements had similar properties.
- 4** The modern Periodic Table arranges elements in order of increasing:
 - A** atomic mass, just as the original one did.
 - B** densities of the elements.
 - C** atomic number, which is the number of protons in the nucleus of each atom.
 - D** atomic number, which is the combined number of protons and neutrons in the nucleus of each atom.
- 5** In the modern Periodic Table, the metallic elements tend to be found:
 - A** towards the left side of the table.
 - B** towards the right side of the table.
 - C** only at the top of the table.
 - D** scattered equally throughout the table.
- 6** The modern organisation of elements in the Periodic Table puts elements which have similar chemical properties:
 - A** in horizontal rows across the table.
 - B** in vertical columns down the table.
 - C** into vertical columns for the metals and into horizontal rows for the non-metals.
 - D** into vertical columns for the non-metals and into horizontal rows for the metals.
- 7** A true statement about atoms of one particular element is that they:
 - A** are all absolutely identical to one another.
 - B** sometimes have different numbers of protons in their nuclei.
 - C** always have the same number of neutrons in their nuclei.
 - D** always have the same number of protons in their nuclei.

- 8** When an element is written using its symbol and numbers, e.g. helium as ${}^4_2\text{He}$, this is really a short-hand way of showing that each atom of the element has:
- A** 4 electrons and 2 protons in it.
 - B** 2 electrons orbiting the nucleus, which has 4 protons in it.
 - C** 2 protons in the nucleus and a mass of 4 units.
 - D** a total of 4 particles in it, 2 of which are protons and 2 of which are electrons.

- 9** If an atom of helium, written as ${}^4_2\text{He}$, were to lose an electron, then it would be written as:

- A** ${}^4_1\text{He}$.
- B** ${}^3_2\text{He}$.
- C** ${}^3_1\text{He}$.
- D** ${}^4_2\text{He}$, as before.

- 10** If an atom of helium, written as ${}^4_2\text{He}$, were to gain a neutron, it would be written as:

- A** ${}^5_2\text{He}$.
- B** ${}^4_2\text{He}$.
- C** ${}^4_3\text{He}$.
- D** ${}^5_3\text{He}$.

Questions 11–25 refer to the following information. The table below represents a simplified Periodic Table, with important regions marked as they would occur in a more realistic version of the Periodic Table. The regions are not intended to be specific Groups or Periods.

	Region 1	
Region 3		Region 4
	Region 2	

- 11** Which region contains the elements which have the most metallic properties?

- A** 1
- B** 2
- C** 3
- D** 4

- 12** Which region contains the elements which have the most non-metallic properties?

- A** 1
- B** 2
- C** 3
- D** 4

- 13** The properties of the elements would become more metallic as we progress:

- A** across region 1 from right to left and also down region 3.
- B** across region 2 from left to right and also down region 3.
- C** down region 3 and also up region 4.
- D** across both regions 1 and 2 from left to right.

- 14** The most metallic of the elements would be found in the vicinity of the intersection of regions:

- A** 1 and 3.
- B** 3 and 2.
- C** 4 and 2.
- D** 4 and 1.

- 15** The most non-metallic element would be found in the vicinity of the intersection of regions:

- A** 1 and 4.
- B** 3 and 2.
- C** 4 and 2.
- D** 1 and 3.

1|6 As we progress from left to right across regions 1 and 2:

- A** the metallic properties become more pronounced.
- B** the metallic properties become more pronounced in region 1, but not in region 2.
- C** the non-metallic properties become more pronounced in both regions.
- D** the non-metallic properties become more pronounced in region 1, but not in region 2.

1|7 The atomic masses of the elements in the Periodic Table increase as we progress from:

- A** right to left across region 1 and from left to right across region 2.
- B** top to bottom in region 3 and from right to left in region 2.
- C** bottom to top in both regions 3 and 4.
- D** left to right in regions 1 and 2 and from top to bottom in regions 3 and 4.

1|8 The valencies of the elements:

- A** are all the same in regions 1 and 2.
- B** tend to show a pattern of vastly different valencies in region 3.
- C** tend to show a pattern across region 1.
- D** are the same in regions 2 and 4.

1|9 In region 3, the valencies of the elements:

- A** show an increase from top to bottom.
- B** are the same as one another if the elements are in the same vertical column.
- C** are the same as those in region 1.
- D** are commonly 4 or 5.

2|0 Elements towards the centre of this simplified table would tend to:

- A** show some properties of both metals and non-metals.
- B** always show properties of metals.
- C** have a valency of either 1 or 2.
- D** exhibit properties only of non-metals.

2|1 The elements within region 3 of the table:

- A** tend not to react with one another to form binary compounds.
- B** tend to react directly with one another to form diatomic molecules.
- C** tend to be able to react with elements from any region of the table, provided that their valencies are all 1.
- D** tend to have mostly metallic properties, but also some non-metallic properties.

2|2 The elements in region 1 of the table are:

- A** all metals but their valencies need not be exactly the same.
- B** very active metals and so readily form compounds with one another.
- C** only able to react with elements in region 3 of the table.
- D** able to react readily with many of the elements in region 4.

2|3 If an element from the top of region 3 reacts with an element from the top of region 4, then:

- A** they would react in the ratio 1:1 by mass.
- B** the bond between them would most likely be of the covalent type.
- C** the bond between them would most likely be of the ionic type.
- D** it would mean that 3.0 g of the first element would react with 6.0 g of the second element.

24 If an element from the top of region 4 reacts with another element from region 4, then:

- A** the bond between them would always be ionic.
- B** the bond formed would be covalent.
- C** equal masses of the two elements would react, since they are in the same region.
- D** it would be difficult to predict what type of bond would form.

25 Some molecules are diatomic and only involve one element. Elements which show this feature:

- A** are nearly always from region 3.
- B** are spread equally over all of the regions.
- C** usually occur anywhere in region 2.
- D** are most commonly found in region 4.

Questions 26–30 refer to the following information.
A more realistic version of the modern Periodic Table is shown below with 8 vertical columns called Groups and horizontal rows called Periods.
Note that this is not a complete table.

Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group VIII

26 The reason for there being 8 groups in a period is that:

- A** there are 8 different kinds of active elements.
- B** there are 8 types of non-metals known.
- C** the properties of elements in the Periodic Table appear to repeat after each set of 8.
- D** the atomic masses of the elements increase in multiples of 8.

27 The most unreactive elements occur in:

- A** the first period.
- B** Group I.
- C** Group VII.
- D** Group VIII.

28 Elements which are considered to be very active occur in:

- A** Groups I and VII.
- B** Group I only.
- C** Groups IV and V.
- D** the first and second periods.

29 The names (i) ‘alkali metals’ and (ii) ‘halogens’ are those which are given to, respectively:

- A** the elements in Groups I and VII.
- B** the elements in Group I and the first period.
- C** any of the elements which react to form ionic bonds.
- D** Group I and Group VIII.

30 The elements which commonly exhibit a valency of 2:

- A** always come from Group II.
- B** usually occur in Groups II and VI.
- C** are to be found only in the second period and Group II.
- D** nearly always come from Group VI.

Questions 31–41 refer to the following information. The table below represents a small section of the Periodic Table. It shows the relative positions of 9 elements in successive Groups (vertical) and Periods (horizontal), but not the exact number of the Groups and Periods to which the elements belong.

1	2	3
4	5	6
7	8	9

In the following questions, you are asked to make some predictions based on the relative positions of the 9 elements.

31 The most ‘electropositive’ element in this table is:

- ☐ A 1.
- ☐ B 7.
- ☐ C 3.
- ☐ D 9.

32 The most ‘electronegative’ element in this table is:

- ☐ A 1.
- ☐ B 7.
- ☐ C 3.
- ☐ D 9.

33 Element 5 would show more metallic properties than the elements numbered:

- ☐ A 6 and 9.
- ☐ B 1 and 4.
- ☐ C 1 and 7.
- ☐ D 2 and 4.

34 The element which would show the most non-metallic properties is numbered:

- ☐ A 9.
- ☐ B 6.
- ☐ C 7.
- ☐ D 3.

35 Two elements which would show the most similarities in properties are elements:

- ☐ A 1 and 3.
- ☐ B 5 and 8.
- ☐ C 3 and 7.
- ☐ D 1 and 9.

36 If an ionic bond were to form between two of the elements then it would most likely be between elements:

- ☐ A 2 and 8.
- ☐ B 7 and 3.
- ☐ C 3 and 9.
- ☐ D 3 and 6.

37 The element which is most likely to be a gas at room temperature is element:

- ☐ A 1.
- ☐ B 7.
- ☐ C 9.
- ☐ D 3.

38 The element which would have the lowest atomic mass is probably element:

- ☐ A 1.
- ☐ B 3.
- ☐ C 7.
- ☐ D 9.

39 The element which would definitely have the greatest atomic number is:

- A** 1.
- B** 3.
- C** 7.
- D** 9.

40 If element 2 were known to have a valency of 4, then the correct valencies of some other elements could well be:

- A** valency 3 for elements 7 and 8.
- B** valency 4 for elements 1 and 3.
- C** valency 3 for elements 4 and 6.
- D** valency 5 for elements 7 and 9.

41 If element 7 had a greater atomic mass than element 8, then it would mean that:

- A** element 7 and element 8 should be changed around.
- B** element 7 has more neutrons in its atoms than element 8.
- C** element 7 has more electrons in its atoms than element 8.
- D** all of the above statements are true.

42 It is true to say that the elements in Group VIII of the Periodic Table:

- A** always have a valency of 8.
- B** are also referred to as 'Group 0' because the valency of these elements is, basically, 0.
- C** are the most active non-metals of all.
- D** were discovered very early because they are active elements.

43 The elements of Group I of the Periodic Table are also referred to as the:

- A** 'earth metals' because they are found within the Earth.
- B** 'alkaline earth' elements since they only form alkalis in solution.
- C** 'noble earth' metals since they are the most active metals.
- D** 'alkali metals' since they always form alkalis when placed in water.

44 Nitrogen is an element which is found in Group V of the Periodic Table. This would mean that it:

- A** could probably show valencies of both 5 and 3.
- B** is much more reactive than Group VII elements.
- C** could only ever have a valency of 5.
- D** could readily form compounds with Group VIII elements.

45 Some elements, such as phosphorus, show two different valencies, depending on the particular compound of which they are part. Such elements:

- A** usually occur across the second period only of the Periodic Table.
- B** are nearly always the elements which are solids at room temperature.
- C** are more commonly non-metals than metals.
- D** occur mainly in Groups I and II.

- 1** When an element such as sodium reacts with an element such as chlorine, the two elements:
 - A** react in any proportions with no 'left-overs'.
 - B** sometimes form a compound and sometimes form a mixture.
 - C** react with one another in very definite proportions by mass.
 - D** only react if there is three times the amount of chlorine as there is sodium.
- 2** When sodium reacts with chlorine and the formula of the resulting compound is written as NaCl it means that:
 - A** 1.0 g of sodium reacted with 1.0 g of chlorine.
 - B** 1 atom of sodium reacted with 1 atom of chlorine.
 - C** 1 molecule of sodium reacted with 1 molecule of chlorine.
 - D** the formula NaCl simply shows the elements present and has nothing to do with amounts in the reaction.
- 3** The formula NaCl suggests that the bond formed between sodium and chlorine is probably:
 - A** an ionic bond since sodium is a metal and chlorine is a non-metal.
 - B** covalent because chlorine forms mainly covalent bonds.
 - C** a very weak bond since it involves a metal and a non-metal.
 - D** an ionic bond, but could be a covalent one, depending on how the NaCl was formed.
- 4** In any chemical reaction between two elements, each element shows a 'combining power' or valency which:
 - A** is usually 1 or 2.
 - B** is usually 1, 2 or 3 but never greater than 3.
 - C** varies depending on the masses of the elements involved.
 - D** is fairly characteristic for each element.
- 5** Whatever number is given to an element's valency (e.g. 1, 2, 3) it is a measure of the number of:
 - A** hydrogen atoms which one atom of the element could combine with.
 - B** oxygen atoms which one atom of the element could combine with.
 - C** grams of a second element which could combine with 1.0 gram of the first.
 - D** molecules of hydrogen which 1 molecule of the first element could combine with.
- 6** If an element such as calcium is known to have a valency of 2, then a single calcium atom would be expected to combine with:
 - A** 2 atoms of the element oxygen.
 - B** 2 atoms of the element hydrogen.
 - C** any other metal with a valency of 2.
 - D** all of the above.
- 7** If an element has a valency of 1 then it would combine with:
 - A** any other element of the same valency.
 - B** many other elements, regardless of their valencies.
 - C** only metallic elements of valency 1.
 - D** only non-metallic elements of valency 1.

8 The ability of two elements to combine with one another is:

- A** really not related to their valencies, but more to their activities and positions in the Periodic Table.
- B** determined by their valencies which have to be reasonably close.
- C** dependent upon their properties, since one has to be a metal and the other a non-metal.
- D** purely a function of how reactive they are and nothing else.

9 If 2.0 g of element X react exactly with 2.0 g of element Y (i.e. without left-overs), then it would mean that:

- A** 2 atoms of X react with 2 atoms of Y.
- B** 6 atoms of X react with 6 atoms of Y.
- C** the valency of both elements is 2.
- D** 6 g of element X would react with 6 g of element Y.

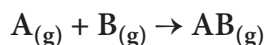
10 When a chemical equation is written to represent a reaction between two elements, it is called an 'equation' because:

- A** the number of atoms of the elements involved before the reaction is equal to the number of atoms after the reaction.
- B** the total mass of the reacting elements would be the same as the total mass of whatever products are formed.
- C** both A and B above are true.
- D** the reaction produces an equal number of products to the number of elements which react.

11 When the element hydrogen reacts with the element oxygen, the formula of the resulting compound is written as H_2O . This formula means that:

- A** there would be twice as many hydrogen atoms as oxygen atoms in any sample of the compound H_2O .
- B** the reaction involved 2 g of hydrogen for every 1 g of oxygen.
- C** hydrogen is twice as reactive as oxygen.
- D** oxygen has a valency less than that of hydrogen and so it is written last.

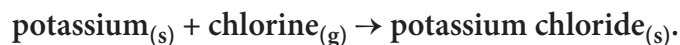
12 If an equation for a chemical reaction is written as:



it would mean that:

- A** both the reactants and the product are gases.
- B** there are two reactants involved, but only one product.
- C** no solids or liquids were involved in the reaction.
- D** all of the above statements are true.

13 A reaction between potassium and chlorine has been written as:



For this particular reaction, it would be true to say that:

- A** the two reactants would combine very readily with one another.
- B** it would be difficult to perform because these two elements usually do not react.
- C** the reaction actually produces a gas, not a solid, at room temperature.
- D** the reason these two elements react with one another is that they both have a valency of 1.

14 When a non-metallic element such as sulfur reacts with another non-metallic element such as chlorine, the bond between the two elements is:

- A** an ionic one.
- B** a covalent one.
- C** extremely weak because the two elements are non-metals.
- D** similar to the bond formed when sodium reacts with chlorine.

15 The reason that sodium atoms do not combine with potassium atoms is that:

- A** the two types of atoms have different valencies.
- B** sodium and potassium have very similar atomic masses.
- C** the two types of atoms are so reactive that they simply could not combine with one another.
- D** they are both electropositive elements from Group I of the Periodic Table and do not tend to combine with other electropositive elements.

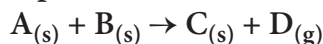
16 It would be true to say that Group II elements such as calcium and barium:

- A** sometimes combine with one another.
- B** don't combine with one another, but readily combine with most non-metallic elements.
- C** are fairly inactive elements.
- D** are active elements, but only combine with elements of Group VI of the Periodic Table.

17 The element carbon, which is positioned towards the middle of a period in the Periodic Table:

- A** is positioned there because it is very unreactive.
- B** shows none of the properties of either metals or non-metals.
- C** combines with many metals and non-metals.
- D** is more reactive than any of the elements near it in the Periodic Table.

18 If a chemical reaction is correctly represented as:



then it would mean that if 4.0 g of $A_{(s)}$ reacted exactly with 5.0 g of $B_{(s)}$:

- A** 4.0 g of $C_{(s)}$ and 5.0 g of $D_{(g)}$ would be formed.
- B** 5.0 g of $C_{(s)}$ and 4.0 g of $D_{(g)}$ would be formed.
- C** 9.0 g of $C_{(s)}$ would be formed since $D_{(g)}$ would have no mass.
- D** the total mass of $C_{(s)}$ and $D_{(g)}$ would be 9.0 g.

19 The relationship

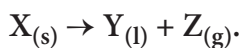
'Total mass of reactants = Total mass of products':

- A** is true for all chemical reactions.
- B** may be true for most chemical reactions.
- C** is only true for those chemical reactions where no energy is gained.
- D** is only true for those chemical reactions where no energy is lost.

20 In chemical reactions:

- A** the number of products is always the same as the number of reactants.
- B** the number of products is not necessarily related to the number of reactants.
- C** there cannot be a greater number of reactants than products.
- D** there cannot be a greater number of products than reactants.

21 A chemical reaction is written as:



Such a reaction is:

- A** unlikely, since there is only one reactant but two products.
- B** very unlikely since a solid reactant would not form a liquid product *and* a gaseous product.
- C** quite possible since the three states of matter could be represented in a single reaction.
- D** not likely because the masses of the products would be greater than that of the reactant.

22 Two chemicals, both solids at room temperature, are known to react with one another. However, when they are in powdered form and mixed thoroughly, they show almost no sign of reacting. This is because:

- A** powders don't generally react with one another.
- B** the powdered forms show different chemical properties to the normal solids.
- C** for powdered forms of chemicals to react, heat always has to be applied.
- D** even in powder form, there is not enough mixing of the chemicals to produce a proper reaction.

23 Reactions between gases generally proceed very well once they are started. This is because:

- A** the molecules of the gases involved are in constant motion and so mix readily.
- B** gases are all very reactive.
- C** they nearly always produce solids and this speeds up the reaction process.
- D** the extra energy which gases possess enables them to react at all times.

24 Solids can be made to mix thoroughly with one another by dissolving them in water so that the particles can move about. This works only if the solids:

- A** are soluble.
- B** don't react with water.
- C** have the properties in both A and B.
- D** are compounds of the heavy metals such as silver and lead.

25 Quite a few elements form compounds which do not dissolve in water. The elements concerned are usually:

- A** from Group I of the Periodic Table.
- B** the very heavy elements such as silver, lead and mercury.
- C** from the non-metallic side of the Periodic Table.
- D** those which form ionic compounds.

26 If chlorine were to combine with oxygen, then a likely formula for the product would be:

- A** ClO.
- B** OCl₂.
- C** O₂Cl.
- D** none of the above.

27 As the reaction between hydrogen and oxygen to produce water proceeds, a lot of energy is given out. That is:
 $\text{hydrogen}_{(g)} + \text{oxygen}_{(g)} \rightarrow \text{water}_{(l)} + \text{energy}$.
 The reason for this is that:

- ☐ A there is less energy stored in the product than was stored in the reactants.
- ☐ B there is more energy stored in the product than was stored in the reactants.
- ☐ C the reaction involves a small loss of mass and this is the source of the energy given out.
- ☐ D energy is needed to start the reaction and it is this energy which is given out as the reaction proceeds.

28 If a chemical reaction has to have energy continually supplied to keep it going then:

- ☐ A it would mean that there must be a greater number of products than reactants.
- ☐ B the products must store more energy than the reactants.
- ☐ C it would certainly not be one representing the burning of a fuel.
- ☐ D B and C would both be true.

29 The terms which are used to describe chemical reactions which (a) absorb energy as they proceed and (b) give out energy as they proceed are, respectively:

- ☐ A (a) endothermic and (b) exothermic.
- ☐ B (a) thermal and (b) thermic.
- ☐ C (a) exothermic and (b) endothermic.
- ☐ D (a) heat producing and (b) heat absorbing.

30 Fuels such as coal, natural gas and petrol burn to give energy. This means that the combustion process would be:

- ☐ A an endothermic reaction on all occasions.
- ☐ B exothermic, but only for large amounts of fuel.
- ☐ C an exothermic reaction on all occasions.
- ☐ D either endothermic or exothermic, depending on the particular fuel.

31 Simple chemical reactions such as those involved in cooking a piece of meat, toasting a slice of bread or cooking a cake are examples of reactions which:

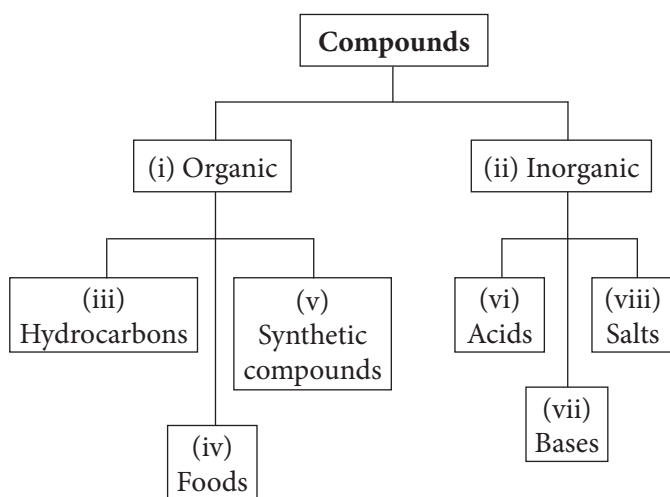
- ☐ A are always exothermic.
- ☐ B are always endothermic.
- ☐ C can be either exothermic or endothermic, depending on the particular reaction.
- ☐ D can be either exothermic or endothermic, depending on the amounts of reactants used.

32 Two elements which occur in the atmosphere are oxygen and nitrogen. However, the amount of nitrogen oxide in the atmosphere is very, very low. This is most likely because:

- ☐ A both gaseous elements are extremely inactive.
- ☐ B oxygen is an extremely inactive gas.
- ☐ C nitrogen is an extremely inactive gas.
- ☐ D carbon dioxide in the atmosphere prevents the two gases from reacting.

- 1** The need to classify chemicals into different groups:
 - A** is unnecessary if chemicals are simply listed in alphabetical order.
 - B** comes about because there are so many (millions!) of chemicals.
 - C** reflects the history of their discovery.
 - D** is based on the relative importance of the chemicals.
- 2** As chemicals are classified into different groups, the important principle is that the chemicals within a particular group:
 - A** have similar physical properties, such as mass and colour.
 - B** are usually in the same state, i.e. all liquids or all solids.
 - C** have chemical formulae which start with the same element, e.g. HCl and H₂O.
 - D** have many chemical properties in common.
- 3** One broad classification of chemicals is to put them into two groups called ‘organic’ and ‘inorganic’. The original classification into these groups generally meant that:
 - A** organic chemicals were those found in living things, while inorganic chemicals were nothing to do with living things and were found within the rocks and the atmosphere of the Earth.
 - B** only food products were included in the group called ‘organic’.
 - C** only minerals such as iron ore and aluminium ore were classified as inorganic chemicals.
 - D** the points in B and C were the critical ones.
- 4** A simple classification of chemicals is to put them into two groups called ‘elements’ and ‘compounds’. When this system is used:
 - A** the ‘compounds’ group would have many more chemicals in it than the ‘elements’ group.
 - B** the ‘elements’ group would have many more chemicals in it than the ‘compounds’ group.
 - C** the group called ‘organic’ chemicals would be left out.
 - D** most of the ‘inorganic’ chemicals would not be included.
- 5** The group of chemicals which we call ‘compounds’ all consist of:
 - A** two elements combined together.
 - B** either two or three elements combined.
 - C** a number of different elements combined.
 - D** a metal and a non-metal combined.
- 6** The chemicals we call mixtures are usually put in a totally different category to compounds when classifying chemicals. This is because:
 - A** there is not a big variety of mixtures.
 - B** mixtures simply do not meet the requirements of the definition of a compound.
 - C** so many mixtures exist that they are difficult to classify.
 - D** mixtures always contain a variety of compounds.

Questions 7–13 refer to the following information.
A key for the classification of chemical compounds is shown below.



7 From the key, it would be true to say that compounds such as nylon, rayon and teflon would:

- ☐ A all be classified as inorganic in category (v).
- ☐ B all be classified as organic in category (v).
- ☐ C be unable to be categorised with this small amount of information.
- ☐ D most likely be classified as salts.

8 The group labelled ‘(iv) Foods’:

- ☐ A would always be organic.
- ☐ B would not contain any ‘Synthetic compounds’.
- ☐ C could have the statements in both A and B correctly applied to it.
- ☐ D should include the organic group ‘(iii) Hydrocarbons’ as well.

9 The key also means that:

- ☐ A hydrocarbons are not inorganic compounds.
- ☐ B acids are generally classified as inorganic compounds.
- ☐ C synthetic compounds are mainly organic.
- ☐ D all of the above statements are true.

10 Keys such as this one which are used in organising and classifying chemical compounds are helpful, but not always perfect. A common reason for this is that:

- ☐ A some compounds might be able to be classified into two different categories.
- ☐ B such keys can miss out on certain groups of compounds like mixtures.
- ☐ C the major classifications are commonly incorrect.
- ☐ D keys ignore how the chemical compounds are made.

11 Citric acid is a compound which occurs naturally in citrus fruits. Trying to classify citric acid creates a problem, since this compound:

- ☐ A is really a synthetic one.
- ☐ B only occurs in very small amounts.
- ☐ C is actually a salt, even though it is called an acid.
- ☐ D is an organic compound, yet it is also an acid.

12 If citric acid were to be classified using the key supplied, then it would:

- ☐ A definitely be put into the ‘acid’ section.
- ☐ B probably be put into the ‘hydrocarbon’ section since it is known to contain both carbon and hydrogen.
- ☐ C have to go into the ‘organic’ section since the first choice is either organic or inorganic.
- ☐ D be put into the ‘synthetic compounds’ section because plants actually do synthesise it.

13 Another problem with this key is that:

- ☐ A the majority of foods are inorganic.
- ☐ B some synthetic compounds, e.g. many silicon ones, are not organic.
- ☐ C many hydrocarbons are inorganic.
- ☐ D all of the above statements are true.

14 Another category which can be used in the classification of compounds is ‘binary compounds’.

1. Carbon dioxide, CO_2
2. Water, H_2O
3. Sulfuric acid, H_2SO_4
4. Octane, C_8H_{18}
5. Glucose, $\text{C}_6\text{H}_{12}\text{O}_6$

Using this classification, the compounds listed above which could be called ‘binary’ are:

- ☐ A 1, 2, 4 and 5.
- ☐ B 1, 2 and 4 since they each have two elements in them.
- ☐ C 1 and 3 since they each have two parts to their names.
- ☐ D 2, 4 and 5.

15 Early classification of acids was based on the common behaviour of the compounds in simple chemical reactions, when little was known about the elements of which they are composed. Such indicators of acids would have been that they:

- ☐ A generally react with metals such as iron and zinc.
- ☐ B are corrosive to the skin.
- ☐ C readily react with compounds called bases.
- ☐ D have all of the properties above.

16 Early investigations into why acids behave in a similar manner suggested that the element oxygen was common to all acids. This was most likely because:

- ☐ A oxygen *is* common to all acids.
- ☐ B oxygen is present in such common acids as sulfuric, nitric, carbonic and phosphoric, which have all been known for a long time.
- ☐ C the presence of the active element oxygen would explain the active nature of all acids.
- ☐ D acids always react to form an oxide.

17 The element which was eventually discovered to be present in all acids and hence the reason for them being similar chemically is now known to be:

- ☐ A hydrogen.
- ☐ B oxygen.
- ☐ C sulfur.
- ☐ D carbon.

18 Modern atomic theory suggests that the property which all acids have in common is that they:

- ☐ A all produce H^+ when put into water.
- ☐ B produce hydrogen gas when they react with heavy metals such as silver and mercury.
- ☐ C react with all metals and produce water.
- ☐ D exhibit all of the above properties.

19 If an acid is said to be ‘weak’, it means that:

- ☐ A there is not much hydrogen in it.
- ☐ B there is not much hydrogen or oxygen in it.
- ☐ C it doesn’t easily form a lot of H^+ ions when put into water.
- ☐ D it has been diluted too much.

20 In the following table, which alternative shows reactions that best indicate the way acids behave?

	Reaction with active metal	Reaction with CaCO_3	Reaction with MgO
A	Do react readily	React, forming CO_2	React readily to form a salt
B	Do react readily	No reaction	No reaction
C	No reaction	Slight reaction	Extremely exothermic reactions
D	No reaction	No reaction	React readily to form a salt

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D D

21 The compounds called ‘bases’ are ones that:

- A** are similar to acids.
- B** are often in the form of solids at room temperature.
- C** act quite differently to acids in most reactions.
- D** have the properties of both B and C.

22 Most compounds classified as ‘bases’:

- A** have non-metals such as phosphorus or sulfur in them.
- B** contain a metal.
- C** rarely contain a metal.
- D** are usually chlorides of metals.

23 A sample of a compound known to be a base is put into water and it dissolves completely. The resulting solution:

- A** would have lost some of the chemical properties of the base in the dissolving process.
- B** would now behave fairly similarly to an acid solution.
- C** is now correctly called an ‘alkali’.
- D** could not be expected to be as effective in reactions as the solid.

24 The chemical feature that all alkalis have in common is that:

- A** they are all metallic oxides.
- B** they all react at the same rate with other reactants.
- C** when they are put into water, they form hydroxide ions, i.e. OH^- ions.
- D** unlike acids, they are non-corrosive compounds.

25 Three common bases are:

- soda – formula Na_2O
- potash – formula K_2O
- lime – formula CaO .

These formulae suggest that:

- A** if they are alkalis, the fact that they form OH^- ions means that they must obtain H from the water.
- B** it is unlikely that they could dissolve to form alkalis.
- C** it is only the active metals that can form bases.
- D** all of the above are true.

26 A feature of acids and bases is that:

- A** they all contain the element ‘oxygen’ as an essential component.
- B** they are capable of reacting together to neutralise the effects of one another.
- C** no foods contain them as they are all extremely toxic.
- D** they are all synthesised from common metallic elements.

27 Acidic and basic compounds occur naturally in foods. Individually, they give distinctive tastes to the foods in which they occur. These tastes are, respectively:

- A** sour and bitter.
- B** bitter and sour.
- C** sweet and sour.
- D** bitter and sweet.

28 The category ‘chemical salts’ refers to a group of compounds which:

- A** are formed when an acid reacts with a base.
- B** do not react like either acids or bases.
- C** usually do not form either H^+ or OH^- ions when dissolved in water.
- D** have all of the above properties.

29 The most common chemical salts are compounds containing:

- A** a metal and a single non-metal.
- B** either hydrogen or a metal combined with a non-metal.
- C** a very active metal and a non-metal.
- D** a metal and an acid radical or appropriate non-metal.

30 In general, chemical salts such as sodium chloride which is used as table salt:

- A** are all harmless and have the same taste as table salt.
- B** only ever contain one metal and one non-metal.
- C** can mostly be made by a reaction between a suitable metal and an appropriate acid.
- D** are all soluble substances, like the acids and alkalis from which they may be formed.

31 The formulae of some common acids, alkalis and salts are listed below.

1. HCl
2. NaOH
3. NaCl
4. HNO₃
5. KNO₃
6. CaO
7. NH₄NO₃

The numbers which indicate an acid, an alkali and a salt, in that order, are:

- A** 1, 3 and 5.
- B** 4, 2 and 7.
- C** 4, 7 and 6.
- D** 4, 3 and 7.

32 The modern interpretation of the classification of compounds as ‘organic’ is that such compounds are:

- A** basically any compounds of carbon.
- B** only found in living things.
- C** the synthetic ones which include nylon, ‘Teflon’ and rayon.
- D** compounds of carbon, except for the synthetic ones.

33 When compounds are described as ‘hydrocarbons’, it means that:

- A** they contain a number of elements which include carbon and hydrogen.
- B** they contain only the elements carbon and hydrogen.
- C** they are fuels such as ethanol which contain carbon and hydrogen.
- D** not only are they compounds of carbon and hydrogen, but the C:H ratio is always 1:2.

34 The following table lists the names and formulae of some compounds of carbon and hydrogen.

Compound name	Formula
Propane	C ₃ H ₈
Acetylene	C ₂ H ₂
Benzene	C ₆ H ₆
Butane	C ₄ H ₁₀

The feature of these particular compounds is that:

- A** they are all common fuels.
- B** they all form CO₂ and H₂O when burned.
- C** they all react with oxygen in exothermic reactions.
- D** all of the above statements apply to them.

35 The chemical name ‘carbohydrates’ describes a group of organic compounds that:

- A** contain the elements C, H and O in any proportion.
- B** are basically the same as hydrocarbons.
- C** contain the elements C, H and O and are a source of energy for humans.
- D** usually combine with oxygen in reactions which are always endothermic.

36 The following table lists the names and chemical formulae of some common carbohydrates.

Carbohydrate	Formula
Fructose	$C_6H_{12}O_6$
Glucose	$C_6H_{12}O_6$
Lactose	$C_{12}H_{22}O_{11} \cdot H_2O$
Maltose	$C_{12}H_{22}O_{11} \cdot H_2O$

According to the table, a feature which all of the carbohydrates have in common is that the elements:

- A** carbon, hydrogen and oxygen are in the same proportions.
- B** have formed compounds which have the same basic formulae.
- C** hydrogen and oxygen are in the same proportions as they are in water.
- D** in the compounds indicate that A and C above are both correct.

37 A feature which hydrocarbons and carbohydrates have in common is that:

- A** they all have the same elements present in them.
- B** they produce carbon dioxide and water when completely burnt.
- C** they all provide energy within the body.
- D** they form molecules which have very similar formulae.

38 The group which is classified as ‘foods’:

- A** refers to the compounds sugars, fats and proteins only.
- B** includes only those compounds which supply our bodies with energy.
- C** includes carbohydrates, fats and proteins.
- D** refers to anything that we consume.

39 The compounds called ‘proteins’ are classified in the ‘food’ category since they:

- A** are the primary source of energy in the human body.
- B** have the basic function of storing energy in the body.
- C** provide the necessary chemicals for growth and repair of body tissue.
- D** have all of the properties above.

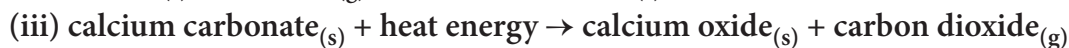
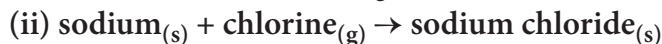
40 The foods in the group called ‘fats’:

- A** are organic compounds which store large amounts of energy.
- B** are not really needed in our diets at all.
- C** can be stored by our bodies, unlike some other food types.
- D** have the features of both A and C above.

41 Vitamins are considered to be foods because they:

- A** are essential for the normal function of our bodies and without them our bodies are prone to certain diseases.
- B** can be ‘burnt’ to release energy, like carbohydrates.
- C** are basic sources of stored energy.
- D** contain only the elements carbon, hydrogen and oxygen as do other carbohydrates.

- 1** Perhaps the simplest type of chemical reaction is a combination reaction. It usually involves elements, or even compounds, bonding together to make a new compound. Three chemical reactions are listed below.



Of the three reactions, the ones which are combination reactions are:

- A** (i), (ii) and (iii). **B** (i) and (ii). **C** (i) and (iii). **D** (ii) and (iii).

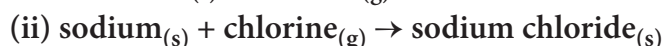
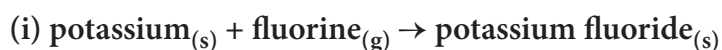
- 2** If a combination reaction does *not* occur between two chemicals, it could mean that:

- A** both of the chemicals are inactive.
B only one of the chemicals is inactive.
C one chemical could have a protective coating, e.g. an oxide, which prevents it from reacting.
D any of the above options might be true.

- 3** When the element helium is mixed with the element argon and the mixture is heated strongly, there is no evidence of a combination reaction occurring. This is due to the fact that:

- A** the mixture has not been heated strongly enough to cause a reaction.
B the proportions of helium and argon are incorrect and so no reaction occurs.
C both elements are always extremely unreactive.
D while helium is quite an active element, argon is not.

- 4** Four combination reactions between metals and non-metals which occur quite readily are:



This is because in every case:

- A** there is an active metal and an active non-metal.
B the metal is an active one, even though the non-metal is inactive.
C the non-metal is an active one, even though the metal is inactive.
D the metals are in the solid state, otherwise there would be no reaction.

- 5** When carbon, the main compound of coal, is burnt in air, it:

- A** readily combines with the oxygen in air to form carbon dioxide.
B forms not only carbon dioxide, but also a considerable amount of carbon nitride.
C forms carbon dioxide which has a greater mass than the amount of carbon burnt.
D reacts such that A and C are both true.

- 6** The reaction between iron and sulfur can be represented as:
 $\text{iron}_{(s)} + \text{sulfur}_{(s)} \rightarrow \text{iron sulfide}_{(s)}$.
 This reaction:
- ☐ A could not possibly occur as both reactants and products are solids.
 - ☐ B would be assisted if the iron and sulfur were in the form of small particles and well mixed.
 - ☐ C should have the product relabelled 'iron sulfate'.
 - ☐ D is not really a combination reaction as it requires a lot of heat to make it proceed.
- 7** The elements magnesium and nitrogen can form the compound magnesium nitride (formula Mg_3N_2). However, if an attempt is made to react magnesium in air with the abundant nitrogen in the air, it fails. This is most likely because:
- ☐ A magnesium will react with the more active oxygen in the air in preference to nitrogen.
 - ☐ B magnesium is not a very active metal.
 - ☐ C the air needs to be heated a little to make magnesium nitride form.
 - ☐ D nitrogen in the air is in the form of diatomic molecules.
- 8** In the steel-making process, lime (CaO) is made to react at high temperature with silica (SiO_2), an impurity in iron ore. The product is calcium silicate (CaSiO_3), commonly known as 'slag'. That is:
 $\text{calcium oxide}_{(s)} + \text{silicon dioxide}_{(s)} \rightarrow \text{calcium silicate}_{(s)}$.
 This reaction would:
- ☐ A be a combination reaction since both of the reactants are bonded together to form a single product.
 - ☐ B not correctly be called a combination reaction since no individual free elements are involved.
 - ☐ C be considered a combination reaction only if the temperature was not too high.
 - ☐ D be difficult to achieve since iron would react with lime.
- 9** Chemical reactions referred to as 'decomposition' reactions:
- ☐ A always result in the formation of the individual elements present in the reactant(s).
 - ☐ B occur only with very large organic molecules.
 - ☐ C involve the break-up of compounds to form simpler compounds and/or elements.
 - ☐ D do not include those where electricity is used to break bonds.
- 10** The solid compound lead nitrate decomposes when heated. One of the products is a different solid lead compound, lead oxide. The *best* indication of a decomposition reaction having occurred is:
- ☐ A a change in colour as lead nitrate changes to lead oxide.
 - ☐ B a loss of mass as the solid lead nitrate changes to lead oxide.
 - ☐ C the formation of a gas.
 - ☐ D that the reaction requires a great deal of heat.

11 The reaction to produce lime (CaO) is one which involves the heating of limestone (CaCO₃); i.e.:
 calcium carbonate_(s) + heat → calcium oxide_(s) + carbon dioxide_(g).

This reaction is:

- ☐ A a decomposition reaction because only two products are formed.
- ☐ B a decomposition reaction because the two products formed come solely from the original reactant.
- ☐ C not a decomposition reaction because no element is formed as a product.
- ☐ D possibly a decomposition reaction, but only if some of the oxygen in the products came from the air.

12 Instances where chemical decomposition occurs include:

- ☐ A the breaking down of mercury oxide to form mercury and oxygen.
- ☐ B the breaking down of water, by electricity, to form hydrogen and oxygen.
- ☐ C the extraction of aluminium metal from its ore, bauxite, using electricity.
- ☐ D all of the above.

Questions 13–16 refer to the following information.

The table below lists four chemicals which are solids, and the colour of each one. 2.00 g of each solid was heated strongly in the presence of air, then cooled and reweighed, and its colour after heating was noted.

No.	Chemical	Before heating		After heating	
		Colour	Mass (g)	Mass (g)	Colour
1	potassium nitrate	white	2.00	1.68	white
2	copper	orange/brown	2.00	2.50	black
3	potassium chlorate	white	2.00	1.22	white
4	iron	silver/grey	2.00	2.57	blackish

13 From this information, it appears that ‘combination’ reactions occur when heating the chemicals numbered:

- ☐ A 1 and 3.
- ☐ B 2 and 3.
- ☐ C 2 and 4.
- ☐ D 1, 2, 3 and 4.

14 From this information, it appears that ‘decomposition’ reactions occur when heating the chemicals numbered:

- ☐ A 1 and 3.
- ☐ B 2 and 3.
- ☐ C 2 and 4.
- ☐ D 1, 2, 3 and 4.

15 The situations in the table where the physical property of colour alone suggests that chemical reactions have occurred are with the chemicals numbered:

- ☐ A 1 and 2.
- ☐ B 2 and 3.
- ☐ C 3 and 4.
- ☐ D 2 and 4.

16 The fact that heating potassium chlorate and potassium nitrate did not produce chemicals with a different colour can best be explained by saying that:

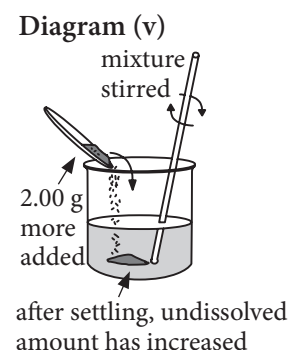
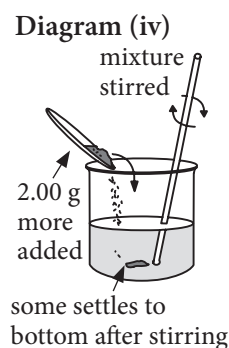
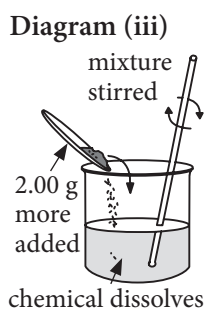
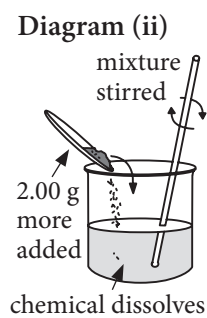
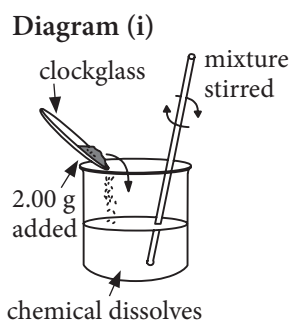
- A** both chemicals did not undergo either combination or decomposition.
- B** the chemicals must have undergone combination reactions.
- C** any gas with which they combined must have been colourless.
- D** all common, simple potassium compounds are white in colour.

17 In chemistry, the meaning of a 'precipitate' is:

- A** a solid forming from a solution.
- B** a substance formed by the reaction of an acid and an alkali.
- C** a gaseous compound which gives colour to a solution.
- D** a solid which is formed when any solution is heated.

Questions 18–21 refer to the following information.

A student carries out a dissolving experiment with a particular compound. She puts 100 mL of water into a beaker. She then adds 2.0 g of the compound at a time, up to 10.0 g, and stirs the solution thoroughly after each addition. The diagrams below show her findings.



18 From the diagrams, it appears that the solution became saturated after the addition of an amount of the chemical:

- A** equal to 2.0 g.
- B** between 2.0 g and 4.0 g.
- C** equal to exactly 6.0 g.
- D** between 6.0 g and 8.0 g.

19 The solubility of the compound is closest to:

- A** 3.0 g/100 mL.
- B** 5.0 g/100 mL.
- C** 7.0 g/100 mL.
- D** 9.0 g/100 mL.

20 Expressed another way, the solubility of the compound is closest to:

- A** 20.0 gL⁻¹.
- B** 40.0 gL⁻¹.
- C** 70.0 gL⁻¹.
- D** 90.0 gL⁻¹.

21 The solid remaining in the bottom of the beaker in diagrams (iv) and (v):

- A** is an example of a precipitate forming from solution.
- B** could not correctly be called a precipitate because it never dissolved and so did not form from solution.
- C** could easily be made to dissolve by cooling the solution.
- D** must be different to the original compound due to a reaction with water.

Questions 22–26 refer to the following information.

The table below lists some soluble compounds and some insoluble ones.

No.	1	2	3	4	5	6	7	8
Insoluble compounds	silver chloride	barium sulfate	calcium carbonate	barium carbonate	silver oxide	silver sulfate		
Soluble compounds	calcium chloride	potassium sulfate	sodium carbonate	barium nitrate	silver nitrate	potassium carbonate	sodium sulfate	calcium nitrate

22 Solid silver chloride could be formed by mixing:

- ☐ A a solution of calcium chloride with silver oxide.
- ☐ B silver sulfate with a solution of calcium chloride.
- ☐ C silver nitrate solution with calcium chloride solution.
- ☐ D both A and B above.

23 If solutions of potassium sulfate and sodium carbonate were mixed, then:

- ☐ A a precipitate of potassium carbonate would form.
- ☐ B a precipitate of sodium sulfate would form.
- ☐ C the carbonate would break up to form carbon dioxide.
- ☐ D no reaction would take place.

24 Barium carbonate solid could be formed by mixing:

- ☐ A solutions of barium nitrate and sodium carbonate.
- ☐ B solutions of potassium carbonate and barium nitrate.
- ☐ C the solutions in both A and B.
- ☐ D barium sulfate with sodium carbonate.

25 The information in the table suggests that the only compound/s of silver that could be formed by precipitation from solutions of the soluble compounds listed:

- ☐ A are silver chloride and silver sulfate.
- ☐ B are silver chloride and silver oxide.
- ☐ C is silver sulfate.
- ☐ D is silver oxide.

26 A true statement which could be made from the information in the table is:

- ☐ A that there are no insoluble compounds of sodium or potassium listed.
- ☐ B that there are no nitrates of metals listed as being soluble.
- ☐ C that carbonates of metals are all soluble.
- ☐ D that chlorides of metals are all insoluble.

27 When solutions of an acid and an alkali are mixed, the resulting reaction is called ‘neutralisation’ because:

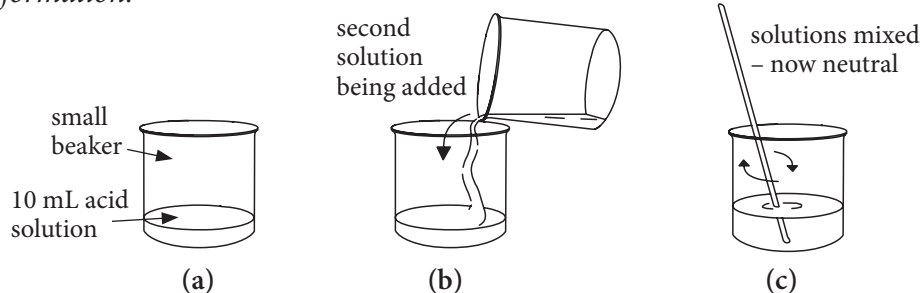
- ☐ A the original and the final solutions are neutral.
- ☐ B the properties of the acid are counter-acted or ‘cancelled out’ by the alkali.
- ☐ C although they mix, there is no reaction between them.
- ☐ D of the reasons in both B and C.

28 A more accurate definition of the chemical reaction called ‘neutralisation’ is that it is one which:

- ☐ A occurs between an acid and an alkali, producing a chemical salt plus water.
- ☐ B occurs between an acid and an alkali, producing a chemical salt and hydrogen gas.
- ☐ C involves the precipitation of a chemical salt.
- ☐ D forms a neutral solution, regardless of what the reactants are.

Questions 29–32 refer to the following information.

A student has 10 mL of acid solution in a beaker and wishes to convert it to a neutral solution by adding the correct amount of a suitable second solution.



29 The *type* of solution she would have to add is:

- ☐ A an acidic one which is weaker than the original.
- ☐ B an acidic one which is stronger than the original.
- ☐ C one which has alkaline properties.
- ☐ D any solution containing a salt.

30 The *amount* of solution she would have to add:

- ☐ A is the same as the amount of solution she had originally, i.e. 10 mL.
- ☐ B has to be 20 mL to overpower the acid’s effect.
- ☐ C depends on whether or not a precipitate formed.
- ☐ D depends on how strong each solution is.

31 When the *exact* amount of the second solution has been added to neutralise the original and the solutions mixed well:

- ☐ A the resulting mixture has neither acidic nor alkaline properties.
- ☐ B a chemical salt would be formed.
- ☐ C a small amount of water would be formed.
- ☐ D all of the above would be true.

32 If the student accidentally added too much of the second solution, then the resulting mixture would:

- A** form more chemical salt than when the exact amounts were mixed.
- B** be alkaline or basic.
- C** be only faintly acidic.
- D** produce no chemical salt at all.

33 If hydrochloric acid solution and the alkali sodium hydroxide solution were reacted so that they exactly neutralised one another, then the products would be:

- A** sodium oxide and water.
- B** sodium chloride and water.
- C** sodium chloride, water and hydrogen gas.
- D** chlorine hydroxide and water.

34 Sometimes an acid solution can react with an alkaline solution to produce a salt which is insoluble. An example is:

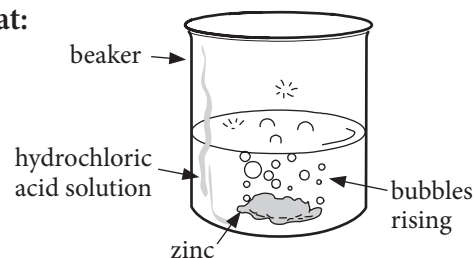


This means that:

- A** the barium sulfate would form as a precipitate.
- B** solid barium sulfate could be obtained from the resulting mixture by filtering and drying it.
- C** this could be a convenient method of making barium sulfate.
- D** all of the above statements are true.

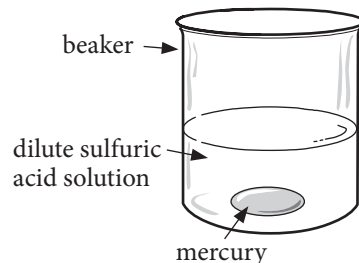
35 When a piece of zinc metal is put into a solution of hydrochloric acid, bubbles of a gas can be seen to rise from the zinc. This means that:

- A** there is definitely a chemical reaction proceeding.
- B** hydrogen gas is being produced by the reaction.
- C** chlorine gas is being produced by the reaction.
- D** both A and B above are true.



36 If some mercury metal is poured into a beaker and completely covered with dilute sulfuric acid, there is no sign of a chemical reaction. This is due to the fact that:

- A** mercury doesn't react with acids because it is a liquid.
- B** the acid solution must be too weak to react with any metal.
- C** mercury is a very inactive metal.
- D** all of the above statements are true.



- 37** Below is an 'Activity Series' of metals with the most active metals on the left and the least active ones on the right. Hydrogen has been placed in the list to show its activity compared to the metals.

K Na Ca Mg Al Zn Fe Sn Pb H Cu Hg Ag Au
 Most active metals $\longrightarrow$ Least active metals
 Decreasing activity

When an acid such as dilute hydrochloric acid or dilute nitric acid is added to samples of each of these metals, the only ones which show a definite reaction are those which are positioned on the left of hydrogen. Those metals on the right of hydrogen show no tendency to react. The reason for this is that:

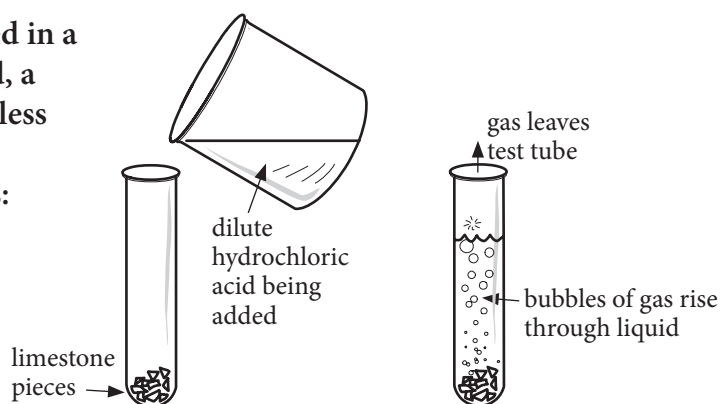
- ☐ A the metals on the left can dissolve in water and so start a reaction.
- ☐ B the metals on the right of hydrogen always have protective coatings which prevent them from reacting.
- ☐ C only metals which are more active than hydrogen can displace it from acid solutions.
- ☐ D all of the metals on the right are too dense to enable a reaction to occur.

- 38** Chemical reactions often follow what is called a 'general pattern', where the products are quite predictable. Such reactions would include the reaction between:

- ☐ A a reasonably active metal and a dilute acid.
- ☐ B an acid and a basic oxide.
- ☐ C an alkali and an acid.
- ☐ D all of the above.

- 39** When small pieces of limestone rock are placed in a test tube and dilute hydrochloric acid is added, a reaction takes place immediately and a colourless gas can be seen to be given off as bubbles rise through the liquid. This gas being produced is:

- ☐ A carbon dioxide and it originates from the limestone.
- ☐ B often hydrogen, which comes from the acid.
- ☐ C sometimes chlorine, which comes from the acid.
- ☐ D oxygen, which comes from the water.



- 40** The reaction in question 39 could correctly be represented as a general pattern reaction because:

- ☐ A limestone, chemical name calcium carbonate, reacts similarly with dilute hydrochloric acid as the other metallic carbonates.
- ☐ B the same gas is produced when calcium metal reacts with dilute hydrochloric acid.
- ☐ C an alkali reacts with limestone in the same way as an acid does.
- ☐ D limestone reacts similarly with most other chemicals.

1 If a chemical reaction is described as one which is 'spontaneous', it means that when the chemicals are mixed, they:

- A** react immediately without any assistance, as in the case of hydrogen and oxygen.
- B** react without any assistance, as in the case of zinc metal and dilute hydrochloric acid.
- C** undergo a vigorous reaction with a gas being given off.
- D** will react vigorously after the addition of a little heat energy.

2 In coal mines, explosions have occurred when the gas methane which is usually trapped in the coal has been released and becomes mixed with air. This means that:

- A** the reaction between methane and air is a spontaneous one.
- B** methane may react with any of the components in the air.
- C** the reaction is not a spontaneous one since methane and air can be mixed without an explosion occurring.
- D** methane gas is extremely reactive with most common reagents.

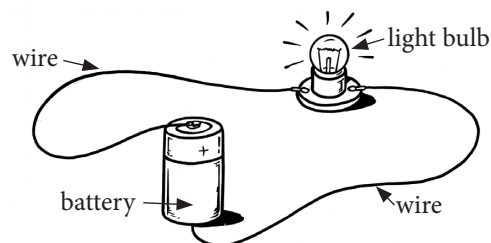
3 The chemical reactions which are called decomposition reactions, such as those involved in extracting a metal from an ore, are:

- A** rarely spontaneous as most of them need the addition of energy to make them proceed.
- B** mostly spontaneous because they usually form elements.
- C** mostly spontaneous except for a few which need heat to start them off.
- D** spontaneous when they involve a metal compound.

4 Examples of spontaneous reactions in chemistry include:

- A** active metals reacting with common dilute acids.
- B** non-metals such as sulfur reacting with metals such as iron.
- C** any reaction between two non-metallic gases.
- D** all of the above.

5 When two wires and a light globe are connected to a battery as shown, electricity flows from one terminal of the battery to the other and lights up the bulb.



This is because:

- A** heat energy in the battery forces electricity through the light bulb.
- B** spontaneous reactions occur in the battery, producing electrical energy which flows through the light bulb.
- C** electricity in the battery causes non-spontaneous reactions to occur.
- D** spontaneous reactions in the battery are reversed, producing electricity.

6 The recharging of rechargeable batteries is an example of:

- A** a spontaneous reaction.
- B** a reaction which needs a lot of heat energy and electrical energy to make it proceed.
- C** a non-spontaneous reaction since it needs the input of energy to make it proceed.
- D** either a spontaneous or a non-spontaneous reaction, depending on the particular battery.

1 The use of indicators in chemistry:

- A** is restricted to acid/base reactions.
- B** mostly involves a chemical which can change colour in the presence of another type of chemical and so reveal its presence.
- C** involves only organic chemicals which can change colour.
- D** involves only inorganic chemicals.

2 A common indicator used in the laboratory is phenolphthalein. It is red/violet in alkali but becomes clear in an acid solution. If a student added some clear phenolphthalein to a clear liquid in a beaker and no colour change occurred, it would mean that:

- A** the liquid in the beaker must be acidic.
- B** phenolphthalein is not a suitable indicator in this situation.
- C** the liquid in the beaker is water.
- D** the liquid in the beaker is definitely not a strong alkali.

3 Indicators such as methyl orange, litmus and phenolphthalein are commonly used with acid/base reactions. This is because they:

- A** are readily available and easily made into solutions.
- B** are safe to use in school situations providing that standard precautions are implemented.
- C** have the properties of both A and B above.
- D** all undergo similar colour changes in acid/base reactions.

Questions 4–7 refer to the following information.

		Indicator			
		litmus	methyl orange	phenolphthalein	bromthymol blue
Colour in these solutions	strong acid	red	red	clear	pale yellow
	weak acid	red	orange	clear	pale yellow
	v.weak acid	reddish blue	yellow	clear	pale yellow
	neutral	blueish red	yellow	clear	green
	v. weak alkali	blueish red	yellow	clear	light blue
	weak alkali	blue	yellow	red	light blue
	strong alkali	blue	yellow	violet	deep blue

4 The table shows that:

- A** all indicators change colour at the point when a solution is neutral.
- B** the colour changes of all the indicators easily distinguish between a strong acid and a strong alkali.
- C** phenolphthalein is the best indicator to distinguish between a weak acid and a weak alkali.
- D** if an unknown indicator is placed into a strong acid solution and it turns red, then the indicator must be litmus.

5 The best indicator for distinguishing between a very weak acid and a very weak alkali is:

- A** bromthymol blue.
- B** litmus.
- C** phenolphthalein.
- D** methyl orange.

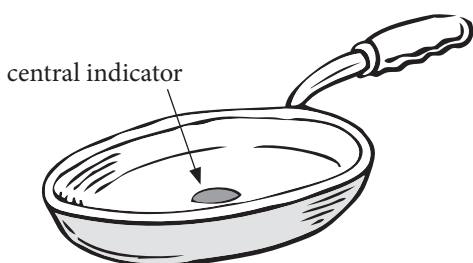
6 Two indicators which would not be useful in distinguishing between a very weak acid and a very weak alkali are:

- A** methyl orange and bromthymol blue.
- B** bromthymol blue and phenolphthalein.
- C** litmus and bromthymol blue.
- D** methyl orange and phenolphthalein.

7 A student slowly adds some alkali solution to an acid solution in a beaker to which a few drops of phenolphthalein has been added. When he stops, the phenolphthalein is colourless or clear. This means that:

- A** he hasn't added enough alkali.
- B** the mixture is still acidic.
- C** the mixture is either still acidic or very weakly alkaline.
- D** the phenolphthalein isn't behaving as the table suggests.

8 Some frying pans have an 'indicator' in the centre which can change colour and also revert to the original colour.



The chemical in this indicator would:

- A** have to be harmless to humans.
- B** probably be responsive to hot and cold conditions.
- C** most likely react with acids and alkalis in food.
- D** have the properties mentioned in both A and B.

Questions 9–10 refer to the following information.

The flowers of some plants change colour depending on the nature of the soil in which they grow. For example, hydrangeas have flowers which are pink in alkaline soil and blue in acid soil.

9 Hydrangea flowers:

- A** must have a chemical in them which acts as an indicator.
- B** could have their colours deliberately changed by the addition of appropriate chemicals to the soil.
- C** have the features of both A and B.
- D** are probably absorbing different metals which change their colours.

10 The majority of plants have a preference for an acidic or alkaline type of soil. However, the acidity or alkalinity must not be too strong or the plants can die. This would imply that if a gardener wanted hydrangeas to have pink flowers, he could safely add a:

- A** slowly dissolving solid basic compound such as calcium oxide (lime).
- B** rapidly dissolving solid basic compound such as sodium hydroxide (caustic soda).
- C** slightly concentrated solution of hydrochloric acid.
- D** very dilute solution of either nitric or sulfuric acid.

11 Special indicators are also used in the detection of:

- A** glucose in urine samples of suspected diabetics.
- B** particular drugs in urine samples of athletes to find 'drug cheats'.
- C** certain compounds in blood samples when diagnosing some diseases.
- D** all of the above.

1 When a chemical equation is described as being 'balanced', it means that:

- ☐ A the number of reactants equals the number of products.
- ☐ B the number of molecules of reactants equals the number of molecules of products.
- ☐ C the total mass of the reactants is equal to the total mass of the products.
- ☐ D all of the above statements are true.

2 When balancing a chemical equation:

- ☐ A the mass of any gaseous product is too small to be considered.
- ☐ B any heat energy involved must be considered.
- ☐ C the mass of a gaseous product is to be considered only if it is the only product.
- ☐ D all reactants and products, regardless of state, must be considered.

3 The reaction between hydrogen and oxygen is correctly represented by the balanced equation:

- ☐ A $2\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{H}_2\text{O}_{(\text{l})}$
- ☐ B $\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow \text{H}_2\text{O}_{(\text{l})}$
- ☐ C $2\text{H}_{2(\text{g})} + 2\text{O}_{2(\text{g})} \rightarrow 2\text{H}_2\text{O}_{(\text{g})}$
- ☐ D $\frac{1}{2}\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow \text{H}_2\text{O}_{(\text{l})}$

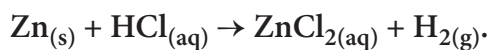
4 A reaction between aluminium metal and dilute hydrochloric acid is best represented by the balanced equation:

- ☐ A $2\text{Al}_{(\text{s})} + 6\text{HCl}_{(\text{aq})} \rightarrow 2\text{AlCl}_{3(\text{aq})} + 3\text{H}_{2(\text{g})}$
- ☐ B $\text{Al}_{(\text{s})} + 3\text{HCl}_{(\text{aq})} \rightarrow \text{AlCl}_{3(\text{aq})} + \text{H}_{2(\text{g})}$
- ☐ C $2\text{Al}_{(\text{s})} + 3\text{HCl}_{(\text{aq})} \rightarrow 5\text{AlCl}_{(\text{aq})} + \text{H}_{(\text{g})}$
- ☐ D $\text{Al}_{(\text{s})} + 4\text{HCl}_{(\text{aq})} \rightarrow \text{AlCl}_{2(\text{aq})} + 2\text{H}_{2(\text{g})}$

5 When balancing equations involving the gases oxygen, hydrogen and chlorine, it is important to remember that:

- ☐ A the mass of hydrogen, the lightest gas known, can be ignored.
- ☐ B each of these gases forms molecules which have 2 atoms in them.
- ☐ C chlorine is not a very active gas.
- ☐ D only the amount of chlorine, the heaviest of the three, needs to be considered.

6 The unbalanced equation for the reaction of zinc metal with dilute hydrochloric acid is:



The coefficients which would correctly balance this equation are, in order from left to right:

- ☐ A 1, 1, 2, 2.
- ☐ B 1, 2, 1, 1.
- ☐ C 2, 4, 2, 2.
- ☐ D both B and C.

7 This is a balanced equation for the reaction between two unnamed chemicals:



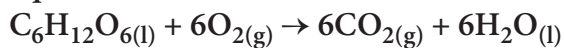
It is found that 50 g of A reacts with 60 g of B and produces 70 g of chemical C. This means that the mass of chemical D produced in the reaction must be:

- ☐ A $3 \times 70 \text{ g}$.
- ☐ B 40 g.
- ☐ C $\frac{1}{3} \times 70 \text{ g}$.
- ☐ D $3 \times 40 \text{ g}$.

8 When a hydrocarbon fuel such as methane gas (CH_4) is burned completely in the atmosphere, it forms carbon dioxide gas (CO_2) and liquid water (H_2O). Since there is no oxygen in methane gas, it can be assumed that the oxygen in water and in carbon dioxide comes from:

- A** another gas which has oxygen in it.
- B** carbon dioxide already in the air.
- C** the atmosphere.
- D** both B and C.

9 If the sugar glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is completely burned in air, it forms carbon dioxide and water according to the equation:



This means that the oxygen in the products:

- A** comes only from the oxygen in the glucose.
- B** is supplied only by the atmosphere.
- C** comes mainly from the atmosphere.
- D** is supplied mainly by the glucose with a small amount coming from the atmosphere.

10 The complete burning of carbon in air produces the gas carbon dioxide. The equation for this reaction and the mass relationships involved are as follows:
Balanced equation: $\text{C}_{(s)} + \text{O}_{2(g)} \rightarrow \text{CO}_{2(g)}$
Mass relationship: $12.0 \text{ g} + 32.0 \text{ g} \rightarrow 44.0 \text{ g}$
These facts mean that, each time 12.0 g of carbon are burned:

- A** 32.0 g of natural oxygen are removed from the air.
- B** there is less oxygen to allow plants and animals to respire.
- C** the amount of carbon dioxide in the air increases by 44.0 g.
- D** all of the above statements are true.

11 According to the information in Question 10, if 1.2 tonnes of carbon in coal were burned completely in air, it would mean that:

- A** 3.2 tonnes of oxygen would be removed from the air.
- B** 4.4 tonnes of carbon dioxide would be put into the air.
- C** the statements in both A and B are correct.
- D** the statements in both A and B are incorrect.

12 When hydrochloric acid reacts with sodium hydroxide, the balanced equation has all of its coefficients equal to 1. That is: $\text{HCl}_{(aq)} + \text{NaOH}_{(aq)} \rightarrow \text{NaCl}_{(aq)} + \text{H}_2\text{O}_{(l)}$
The reason for this is that:

- A** all of the interacting particles have a valency of 1.
- B** acids and alkalis always react in the proportions of 1:1.
- C** oxygen has a valency of 2.
- D** all of the above are true.

13 In the process of extracting iron from its ore, one appropriate reaction can be summarised by this balanced equation:
 $2\text{Fe}_2\text{O}_{3(s)} + 3\text{C}_{(s)} \rightarrow 4\text{Fe}_{(s)} + 3\text{CO}_{2(g)}$
iron ore + carbon $\rightarrow$ iron + carbon dioxide
It would seem that, in this reaction:

- A** more iron is produced than was in the iron ore.
- B** the carbon seems to play the role of getting the oxygen out of the iron ore.
- C** the same mass of carbon dioxide is produced as carbon used.
- D** oxygen would have to be supplied by the air to make the reaction proceed.

1 Some chemical reactions occur very rapidly while others may take extremely long periods of time to complete. An example of (a) a very fast reaction and (b) a very slow reaction would be:

- A** (a) the rusting of iron, (b) the dissolving of iron in a dilute acid.
- B** (a) the reaction between hydrogen and oxygen, (b) the rusting of iron.
- C** (a) the reaction between hydrogen and oxygen, (b) the reaction of sodium metal with water.
- D** (a) the dissolving of zinc in dilute acid, (b) the combustion of a hydrocarbon fuel such as petrol.

2 If any chemical reaction is to proceed at even a moderate rate, then it would be necessary for the reactants to be:

- A** extremely reactive chemicals.
- B** heated vigorously.
- C** mixed fairly thoroughly to allow them to be in contact.
- D** in the gaseous state since this is the only state where the amount of mixing will allow a reaction.

3 If copper (or brass, which contains copper) is exposed to the air, it slowly 'tarnishes' or changes colour with time, even though copper is a fairly inactive metal. This probably means that:

- A** the copper reacts very slowly with some component of the air.
- B** the copper must have some impurities in it for it to behave this way.
- C** acids in the air are responsible, since only they can cause metals to tarnish.
- D** nitrogen in the air reacts with the copper.

4 Iron and sulfur are two solid elements which do react when heated with one another to form the compound iron sulfide. However, if a single solid piece of iron with a volume of about 1cm^3 is placed in sulfur powder and heated, there is only a very small amount of iron sulfide produced. The best way of increasing the yield of iron sulfide would be to:

- A** flatten the piece of iron to make it thinner.
- B** dissolve the iron in a concentrated acid prior to mixing with the sulfur.
- C** dissolve the sulfur in a suitable solvent before mixing with the iron.
- D** have the iron and sulfur in powdered form so that they can mix to a greater extent.

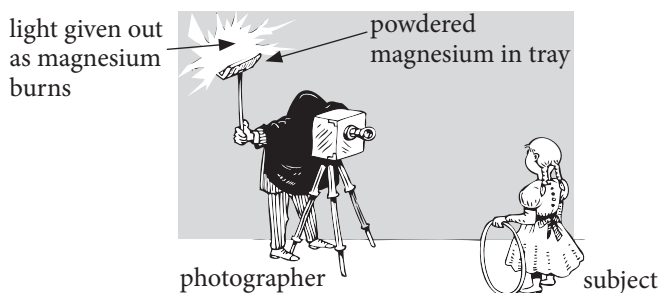
5 Solutions of two different chemicals are known to react with one another. When they are mixed, the simplest way of making sure that there is maximum contact between the reacting particles is to:

- A** stir the mixture.
- B** heat the mixture.
- C** let the solutions mix of their own accord without any interference.
- D** cool the mixture.

6 When an indicator is placed in an acidic or alkaline solution, its rate of reaction with the solution:

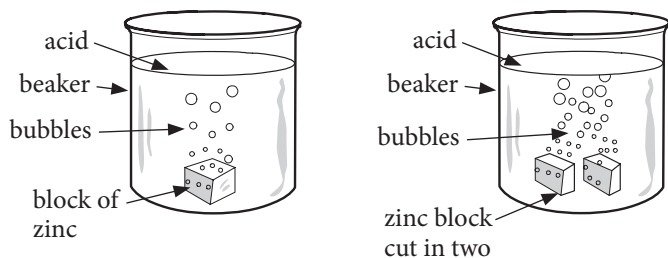
- A** varies greatly with the particular indicator.
- B** depends on whether the solution is acid or alkali.
- C** depends greatly on the concentration of the solution.
- D** is always very fast.

- 7** Magnesium reacts very rapidly with oxygen once it is ignited, and the reaction gives out a great deal of light energy. Magnesium, in a powdered form, was once used in photography to provide light at the instant a photograph was taken.



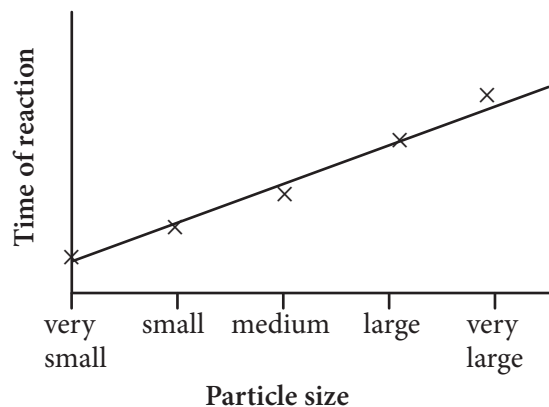
The reason for using magnesium in a powdered form was most likely:

- A** because that is how magnesium is found in nature.
 - B** to allow maximum contact with the air and so produce a rapid reaction when ignited.
 - C** for safety reasons.
 - D** that it doesn't react at all when it is in the form of a single piece of metal.
- 8** Solid zinc metal reacts readily with a dilute acid. If a 1.0 cm^3 block of zinc is placed in dilute hydrochloric acid, bubbles of gas can be seen to form at the surface of the zinc and rise upwards through the liquid.



If a similar block is cut in half and placed in the same acid, it would be expected to react more quickly because:

- 9** In a reaction between zinc and dilute hydrochloric acid, a student uses 5 samples of zinc which have the same mass but are composed of particles of different sizes. She records the 5 sizes as 'very large', 'large', 'medium', 'small' and 'very small'. She puts the zinc sample into excess hydrochloric acid each time and records the time for a reaction to go to completion. She then graphs her results, as shown.



Her graph shows that:

- A** there is no relationship between particle size and time of reaction.
 - B** the larger the particle size, the smaller the time of reaction.
 - C** the smaller the particle size, the smaller the time of reaction.
 - D** there is not enough information to make any real conclusions.
- 10** It is generally accepted that the addition of heat energy:
- A** slows down the majority of chemical reactions.
 - B** speeds up the majority of chemical reactions.
 - C** has little effect on the rate of most chemical reactions.
 - D** is the only factor that determines the rate of a chemical reaction.

11 Outcrops of the rock limestone and limestone statues are slowly ‘eaten away’ in nature as they react with acids in the air and in natural waters. However, when a piece of limestone is put into a dilute acid in the laboratory, it can be ‘eaten away’ quite quickly. The best way of explaining this is to say that:

- A** the naturally occurring acids are much more dilute than those commonly used in the laboratory.
- B** acids in the air are all of very different types to those used in the laboratory.
- C** the limestone used in the laboratory is different to the limestone in nature.
- D** all of the above are true.

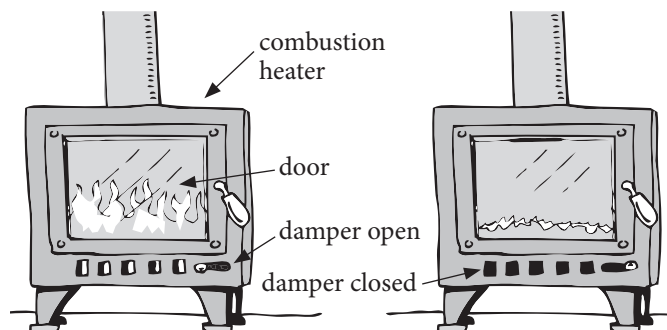
12 With some chemical reactions, additional chemicals called ‘catalysts’ are used. The properties of a catalyst are that it:

- A** speeds up a particular reaction and can be easily seen.
- B** assists the reaction to proceed and can be recovered unaltered at the end of the reaction.
- C** stops the reaction from giving out any heat energy.
- D** can reduce heat production and cause the reaction to produce a different chemical.

13 Catalysts are used in motor vehicles to assist the complete burning of the fuel. The reason for their use is that:

- A** the burning is so quick in an engine that not all of the fuel has time to burn.
- B** the catalysts lessen the amount of pollutants released into the air.
- C** they assist in both of the issues covered by A and B above.
- D** they stop any carbon dioxide escaping into the atmosphere.

14 In a slow combustion heater, the rate of burning of the fuel (usually wood) can be controlled by the use of a ‘damper’. When open, the damper allows a plentiful supply of air into the heater to increase the rate of combustion. When closed, it cuts off the supply of air and decreases the rate of combustion.



The role of the damper is really to:

- A** prevent heat entering the combustion heater.
- B** stop heat from leaving the heater.
- C** prevent the wood from burning too quickly when the wood is damp.
- D** control the amount of oxygen entering the heater and so regulate the rate of burning of the wood.

15 The time it takes for liquid fibreglass to set using a ‘hardener’ (catalyst) would be:

- A** increased if more hardener is used.
- B** decreased if more hardener is used.
- C** not affected by the amount of hardener.
- D** only dependent on the temperature.

- 1** The main reason for processing chemicals is to:
 - A** ensure that they are not poisonous.
 - B** make an end product that has suitable, desired properties.
 - C** lower the cost of making the chemicals.
 - D** make them totally inactive and safe to handle.
- 2** An alloy is a type of mixture, usually of two elemental metals. For example, brass is an alloy of mainly copper and zinc. 'Stainless steel' is an alloy of steel and chromium. The property of stainless steel that is the reason for its manufacture is that it:
 - A** does not oxidise in the air and so does not rust as iron does.
 - B** is extremely malleable and can be shaped more easily than most other metals.
 - C** is much lighter (i.e. less dense) than iron.
 - D** has less iron in it than pure iron and so is cheaper to produce.
- 3** The smell of the interior of a new car is mainly due to chemicals called plasticisers. These chemicals are present in vinyl and 'synthetic leather' upholstery and escape over time. The 'plastic' of a biro or a hard cutting board would not have these plasticisers to such a large degree. The reason for using these plasticisers is probably to:
 - A** give the property of flexibility to the vinyl or plastic.
 - B** ensure that the vinyl or plastic is safe for humans.
 - C** make the plastic or vinyl hard and strong.
 - D** make the plastic or vinyl more dense.
- 4** When margarine is originally made, it is not the colour that we see in the final product. A yellow additive is put into it before it is presented for sale. The most likely reason for this is to:
 - A** make the margarine have a soft texture.
 - B** make the margarine have a more agreeable taste.
 - C** preserve the margarine as yellow additives are the best preservatives.
 - D** give margarine an appearance similar to butter.
- 5** The chemical sulfur dioxide (SO_2) or other chemicals which release sulfur dioxide are commonly added to foodstuffs and recorded as 'additive 220'. Its role in the foodstuffs is to:
 - A** intensify and so improve the flavours in the foods.
 - B** prevent decay of the foods by microbial action and so act as a preservative.
 - C** prevent the 'coagulation' or 'caking' of foods in powder form.
 - D** act as a colouring agent.
- 6** The chemical 'tetraethyl lead' used to be added to petrol to make it perform better in car engines. However, this practice has been phased out because:
 - A** lead has become an element in very short supply.
 - B** it became too costly to keep adding lead compounds to petrol.
 - C** the practice led to the metal lead being released into the atmosphere and lead is a toxic chemical.
 - D** ethanol has been found to do the job as well as tetraethyl lead.

- 1** There are only a few metallic elements which can be found in their 'native' or elemental states in nature. Three such elements include:
- A** iron, aluminium and gold.
 - B** gold, silver and platinum.
 - C** sodium, silver and lead.
 - D** calcium, gold and platinum.
- 2** The elements which are found in their uncombined or elemental states:
- A** are only ever solid metals.
 - B** are often in liquid form.
 - C** include some solid metals and a few gaseous non-metals.
 - D** are seldom found as solids.
- 3** For metallic elements to be found in their 'native' or elemental states, it would mean that the elements:
- A** never react with other elements.
 - B** are very dense and are found at extreme depths in the Earth.
 - C** don't react readily with components of the atmosphere such as oxygen, water and carbon dioxide.
 - D** are probably from Groups II and III of the Periodic Table.
- 4** It is true to say that the term 'ore' most correctly describes:
- A** compounds of metals only.
 - B** compounds of any elements.
 - C** metals which occur in their elemental states.
 - D** the complex compounds of metals which always occur at great depths in the Earth.

- 5** Most common ores are:
- A** relatively simple compounds of metals.
 - B** extremely complex compounds of metals.
 - C** ones from which the metal can be extracted using very little energy.
 - D** very similar in colour and density.

- 6** The table lists the names and chemical formulae of some common ores.

Name of ore	Chemical formula
Galena	PbS
Sphalerite	ZnS
Cassiterite	SnO ₂
Magnetite	Fe ₃ O ₄

The ores from which lead and iron, respectively, could be extracted are:

- A** galena and cassiterite.
 - B** cassiterite and magnetite.
 - C** galena and magnetite.
 - D** sphalerite and magnetite.
- 7** The extraction of most metals from their ores mainly involves:
- A** physical processing since the ores are mixed in with rock material.
 - B** the breaking up of a compound to form the metal.
 - C** the input of energy since the process is nearly always endothermic.
 - D** both B and C above.

- 8** Below is an 'Activity Series' of metals with the most active metals on the left and the least active ones on the right.

K Na Ca Mg Al Zn Fe Sn Pb Cu Hg Ag Au Pt
 Most active metals $\longrightarrow$ Least active metals
 Decreasing activity

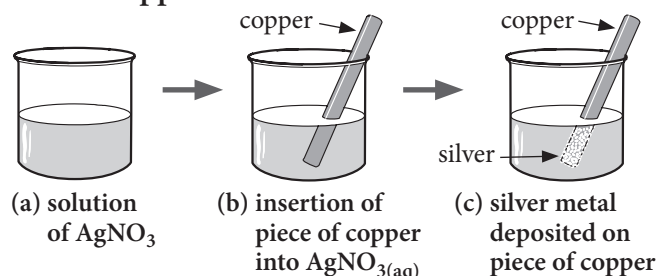
This Activity Series explains why:

- A** potassium, sodium and calcium are never found as elements in nature.
- B** silver and gold are found in nature as elements.
- C** the naturally occurring simple compounds of lead, copper and mercury can be broken down fairly easily to yield the metal.
- D** all of the above statements are true.
- 9** The element gold is sometimes found in riverbeds as small particles, not combined with any other elements. This would best be explained by saying that:
- A** gold is insoluble in water and doesn't react with water.
- B** gold doesn't react with water but can dissolve easily in it.
- C** river water is not quite warm enough to make gold react.
- D** there are no other chemicals available to react with the gold.

- 10** Since sodium and potassium are very reactive elements and form soluble compounds, it probably means that:

- A** their ores are found mainly in arid regions.
- B** the major sources of these elements would be the oceans where their compounds are in dissolved form.
- C** they are to be found as simple compounds, undissolved, in most rocks.
- D** ore deposits of the metals are found in riverbeds.

- 11** A common method of extracting a metal from a solution of its salt is by displacement. For example, silver can be extracted from its solution at the expense of copper:



This method will work if the metal 'inserted' is:

- A** of the same valency as the one in solution.
- B** less active than the metal in solution.
- C** more active than the metal in solution.
- D** very hot at the time it is inserted into the solution.
- 12** According to the 'Activity Series' of the metals in question 8, the displacement method of extraction should work in the situations below listed as:

	A	B	C	D
Metal in solution	Zinc	Copper	Lead	Silver
Metal solid inserted into solution	Copper	Zinc	Silver	Lead

- A** A and B.
- B** B and C.
- C** A and C.
- D** B and D.
- 13** In the displacement method of metal extraction, it would be true to say that:
- A** the metal in solution is in the form of positive ions.
- B** the solid metal which is inserted into the solution is in the form of charged ions.
- C** both of the metals concerned are in the form of unbalanced, charged ions.
- D** both of the metals concerned are in the form of complete, uncharged atoms.

1 People should be concerned about ‘heavy’ metals because, in larger quantities, they are not readily eliminated from the body and can cause health problems. Three such metals are:

- ☐ A sodium, potassium and lithium.
- ☐ B sodium, potassium and lead.
- ☐ C mercury, lead and zinc.
- ☐ D lithium, zinc and calcium.

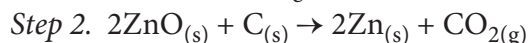
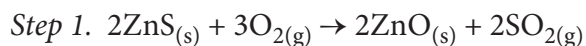
2 In the movie *Erin Brokovich*, it was shown that a particular chromium compound was responsible for horrendous health problems as people unknowingly absorbed the compound in their daily lives. And yet chromium is an essential element for our bodies. The fact that chromium can cause major health problems probably means that:

- ☐ A only very small amounts are needed for the healthy functioning of our bodies.
- ☐ B it is some other metal associated with chromium that causes the health problems.
- ☐ C it is only when the chromium is in a concentrated form and absorbed continually by our bodies that it becomes a health problem.
- ☐ D both A and C are true.

3 A problem with heavy metals, even if they are present in very tiny amounts in living things, is that:

- ☐ A they accumulate via the food chain.
- ☐ B plants tend to accumulate them more than animals.
- ☐ C only carnivorous animals can eliminate them.
- ☐ D they give very bad tastes to food.

4 Industrial plants called ‘smelters’ fuse or melt metal ores in order to extract the metals. For example, to produce zinc, the ore ‘sphalerite’ (ZnS) is roasted in air to produce zinc oxide (ZnO). The oxide is then heated strongly with carbon to produce the metal. The equations for the reactions are:



The effect of these processes would be to:

- ☐ A produce gases which are acidic oxides.
- ☐ B use a lot of oxygen from the air for heating.
- ☐ C use a lot of energy to produce the metal zinc.
- ☐ D do all of the above.

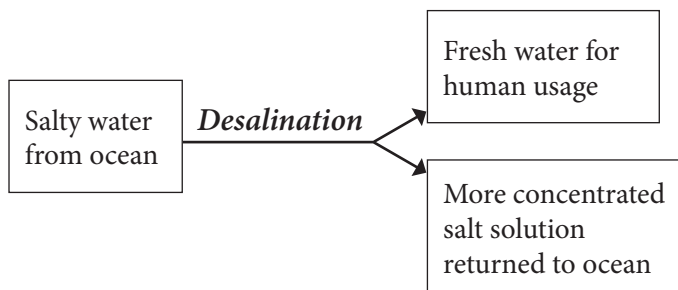
5 A major environmental concern with smelters is that:

- ☐ A because of the high temperatures used, some heavy metals may vaporise and escape into the air.
- ☐ B transportation of the ore is very risky.
- ☐ C both A and B above are true.
- ☐ D they release excessive amounts of oxygen into the air.

6 Because zinc is a solid, it means that as it is processed in a smelter:

- ☐ A there is no chance of it getting into the air.
- ☐ B it is too heavy to get into the air.
- ☐ C it could not pollute waterways.
- ☐ D it could vaporise a little into the air since smelters use very high temperatures.

- 7** When sea water is desalinated to produce fresh water for use by humans, the salt is not extracted as a solid, but is recycled back into the ocean in a more concentrated form of salt water. That is:



An environmental problem arising from this process is that:

- A** the fresh water produced still contains a small amount of salt.
 - B** the concentrated salt solution returned to the ocean is too strong for natural marine communities.
 - C** fresh water from such a source contains higher than normal concentrations of heavy metals.
 - D** delivery of the water to coastal areas is extremely difficult to achieve.
- 8** Combustion of coal puts a lot of carbon dioxide gas into the air. However, it also puts other oxides into the air and these become pollutants because they can cause 'acid rain'. These oxides would be of elements such as:
- A** the non-metals sulfur, nitrogen and phosphorus.
 - B** sulfur, silicon and potassium.
 - C** sodium, potassium and phosphorus.
 - D** the metals sodium, potassium and calcium.

- 9** To prevent overheating, car engines are cooled by a liquid, mainly water, circulating through the engines which are usually made of iron. A chemical called 'anti-freeze' is added to the water so that it will not freeze in very cold conditions. The anti-freeze chemical would have to:

- A** change the freezing point of water to below 0°C.
- B** be one which is not evaporated extremely easily and so lost.
- C** be one which is not corrosive to iron and other metals.
- D** possess all of the above properties.

- 10** Glues and adhesives mostly have a solvent in them to stop the glue from setting before it is used. Once the glue is used, the solvent evaporates into the air and the glue sets. This means that:

- A** the adhesive solvent must be a volatile chemical.
- B** using glues and adhesives puts some pollutants into the air.
- C** both A and B above are true.
- D** the amounts of solvents put into the air are so small that they could never harm the environment.

- 11** The act of pouring unwanted silver nitrate solution down the sink in a laboratory has an effect on the environment which:

- A** can be ignored since it is such a small amount.
- B** creates no problem since it remains in the drains.
- C** is bad because it does find its way back into living systems.
- D** is not an issue because it is soluble.

ANSWERS

Topic 1 – Elements and atomic structure

1 D **2** A **3** B **4** D **5** C
6 A **7** C **8** D **9** B **10** C
11 D **12** D **13** A **14** B **15** C
16 B **17** B **18** D **19** A **20** A
21 A **22** C **23** A **24** C **25** B
26 B **27** C **28** D **29** C **30** B
31 D **32** A **33** B **34** C

Topic 2 – The Periodic Table

1 D **2** B **3** A **4** C **5** A
6 B **7** D **8** C **9** D **10** A
11 C **12** D **13** A **14** B **15** A
16 C **17** D **18** C **19** B **20** A
21 A **22** D **23** C **24** B **25** D
26 C **27** D **28** A **29** A **30** B
31 B **32** C **33** A **34** D **35** B
36 B **37** D **38** A **39** D **40** C
41 B **42** B **43** D **44** A **45** C

Topic 3 – Reactions between elements

1 C **2** B **3** A **4** D **5** A
6 B **7** B **8** A **9** D **10** C
11 A **12** D **13** A **14** B **15** D
16 B **17** C **18** D **19** A **20** B
21 C **22** D **23** A **24** C **25** B
26 B **27** A **28** D **29** A **30** C
31 B **32** C

Topic 4 – Classifying chemicals – names and formulae

1 B **2** D **3** A **4** A **5** C
6 B **7** B **8** C **9** D **10** A
11 D **12** C **13** B **14** B **15** D
16 B **17** A **18** A **19** C **20** A
21 D **22** B **23** C **24** C **25** A
26 B **27** A **28** D **29** D **30** C
31 B **32** A **33** B **34** D **35** C
36 D **37** B **38** C **39** C **40** D
41 A

Topic 5 – Types of chemical reactions

1 B **2** D **3** C **4** A **5** D
6 B **7** A **8** A **9** C **10** B
11 B **12** D **13** C **14** A **15** D
16 D **17** A **18** D **19** C **20** C
21 B **22** C **23** D **24** C **25** A
26 A **27** B **28** A **29** C **30** D
31 D **32** B **33** B **34** D **35** D
36 C **37** C **38** D **39** A **40** A

Topic 6 – Spontaneous and non-spontaneous reactions

1 B **2** C **3** A **4** A **5** B
6 C

Topic 7 – Indicators

1 B **2** D **3** C **4** B **5** A
6 D **7** C **8** D **9** C **10** A
11 D

Topic 8 – Balancing equations

1 C **2** D **3** A **4** A **5** B
6 D **7** B **8** C **9** C **10** D
11 C **12** A **13** B

Topic 9 – Rates of reactions and controlling reactions

1 B **2** C **3** A **4** D **5** A
6 D **7** B **8** C **9** C **10** B
11 A **12** B **13** C **14** D **15** B

Topic 10 – Processing chemicals and desired properties

1 B **2** A **3** A **4** D **5** B
6 C

Topic 11 – Extraction of metals from ores

1 B **2** C **3** C **4** A **5** A
6 C **7** D **8** D **9** A **10** B
11 C **12** D **13** A

Topic 12 – Environmental issues

1 C **2** D **3** A **4** D **5** A
6 D **7** B **8** A **9** D **10** C
11 C

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